Chapter 18

Acute Heart Failure(s)

Movement is the cause of all life.

Leonardo da Vinci

Despite the emphasis on a "heart healthy" lifestyle in modern times, heart failure has grown in prevalence and associated mortality in recent years; i.e., the number of adults in the United States with heart failure increased from 6 million in 2018 to 6.7 million in 2020, and deaths attributed to heart failure were 50% higher in 2020 than in 2010 (1). Current surveys show that one of every 8 deaths in the United States is related to heart failure (1), indicating that heart failure is a leading health problem in this country.

Heart failure is not a single entity (as the title suggests), and is classified according to the portion of the cardiac cycle that is affected (systole or diastole) and the side of the heart that is involved. This chapter describes each type of heart failure, and focuses on the acute, decompensated stage of heart failure that requires management in the hospital. (*Note:* This chapter does not include cardiogenic shock, which is described in Chapter 16.) Many of the recommendations in this chapter are borrowed from the clinical practice guidelines listed at the end of the chapter (2–5).

TYPES OF HEART FAILURE

Heart failure can be the result of impaired cardiac filling (diastolic dysfunction) or a decrease in contractile strength (systolic dysfunction), and these abnormalities can predominate in the left or right side of the heart.

Systolic vs. Diastolic Dysfunction

The influence of systolic and diastolic dysfunction on measures of cardiac performance are shown in Figure 18.1. The upper graph shows the relationship between ventricular end-diastolic pressure (EDP) and stroke volume (which are known as "ventricular function curves"). The curve representing heart failure has a decreased slope, and the point on the curve indicates that heart failure is associated with an increase in EDP and a decrease in stroke volume. The lower graph shows the relationship between ventricular end-diastolic pressure and end-diastolic volume

(EDV). The curve representing diastolic dysfunction has a decreased slope, which reflects a decrease in ventricular compliance (distensibility) according to the following relationship:

$$\text{Compliance} = \Delta EDV/\Delta EDP \tag{18.1}$$

The points on the ventricular compliance curves indicate that the increase in EDP in heart failure is associated with an increase in EDV if the problem is systolic dysfunction, and a decrease in EDV if the problem is diastolic dysfunction. Therefore, *the end-diastolic volume (not the end-diastolic pressure) can distinguish between systolic and diastolic dysfunction in patients with heart failure*. An end-diastolic volume of 97 mL/m^2 (measured relative to body surface area in m^2) is used to distinguish between systolic and diastolic dysfunction i.e., an EDV >97 mL/m^2 indicates systolic dysfunction, and an EDV ≤97 mL/m^2 indicates diastolic dysfunction (6). Because EDV is not easily measured at the bedside, the "ejection fraction" is used to distinguish between systolic and diastolic dysfunction (see next).

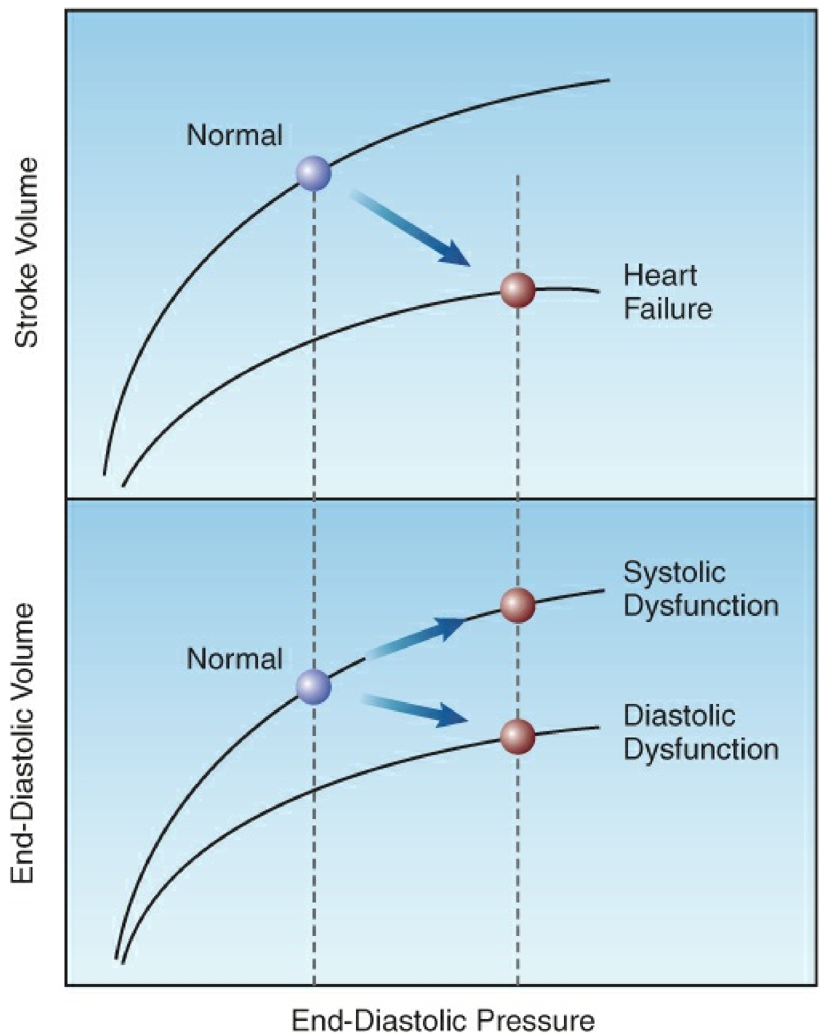

FIGURE 18.1 Graphs showing the influence of systolic and diastolic dysfunction on measures of cardiac performance. Upper panel shows ventricular function curves, and lower panel shows diastolic pressure-volume curves. See text for explanation.

Ejection Fraction

The fraction of the ventricular volume that is ejected during systole is called the *ejection fraction* (EF), and is the ratio of stroke volume (SV) to end-diastolic volume (EDV):

$$EF = (SV/EDV) \times 100 \tag{18.2}$$

(The ratio is multiplied by 100 to express it as a percentage.) The EF is directly related to the strength of ventricular contraction, and is used a measure of systolic function. The normal EF of the left ventricle is ≥55%, but a lower value of ≥50% is used as normal in patients with heart failure because increases in afterload can reduce the EF by 5–10% (2).

The classification of left heart failure based on the left-ventricular ejection fraction (LVEF) is shown in Table 18.1 (2).

. Heart failure caused by diastolic dysfunction has a normal LVEF (≥50%), and is called *heart failure with preserved ejection fraction* (HFpEF).

. Heart failure caused by systolic dysfunction has a reduced LVEF (<50%), and is subdivided into *heart failure with mildly reduced ejection fraction* (HFmrEF) if the LVEF is 41–49%, and *heart failure with reduced ejection fraction* (HFrEF) if the LVEF is ≤40%.

TABLE 18.1 Classification of Heart Failure Based on Left Ventricular Ejection Fraction (LVEF)

Category	LVEF	Problem
Heart Failure with preserved ejection fraction (HFpEF)	≥50%	Diastolic Dysfunction
Heart Failure with mildly reduced ejection fraction (HFmrEF)	41–49%	Systolic Dysfunction (mild)
Heart Failure with reduced ejection fraction (HFrEF)	≤40%	Systolic Dysfunction (moderate/severe).

From the clinical practice guidelines in Reference 2.

Diastolic Dysfunction

Early descriptions of heart failure attributed most cases to systolic dysfunction, but diastolic dysfunction is now recognized as a cause of at least 50% of cases of left heart failure (2). The functional problem with diastolic dysfunction is a decrease in ventricular distensibility that impairs ventricular filling during diastole. The result is a decrease in end-diastolic volume (EDV), and although the EF is preserved, the stroke volume is reduced because the EF is a fraction of a lower EDV.

Common causes of heart failure with preserved EF (HFpEF) include ventricular hypertrophy, myocardial ischemia (stunned myocardium), restrictive (fibrotic) cardiomyopathy, and pericardial tamponade. High intrathoracic pressures during mechanical ventilation can also impair ventricular filling and aggravate HFpEF.

Right Heart Failure

Right-sided heart failure is usually a consequence of pulmonary hypertension (e.g., from pulmonary emboli or chronic lung disease) or inferior wall myocardial infarction. The principal mechanism is systolic dysfunction, and the result is an increase in right-ventricular end-diastolic volume (RVEDV). This can be difficult to detect clinically because the central venous pressure (CVP) does not rise (and jugular venous distension does not occur) until the increase in RVEDV is restricted by the pericardium (*pericardial constraint*) (7).

Echocardiography

Transthoracic echocardiography (TTE) is the standard method for detecting right heart failure at the bedside. An example of a 2D ultrasound image showing right heart failure is shown in Figure 18.2 (7). This is a parasternal short-axis view showing right ventricular enlargement. The right ventricular (RV) cavity is normally about two-thirds the size of the left ventricular (LV) cavity (8), and the image in Figure 18.2 shows that the RV cavity is larger than the LV cavity. Note also that the interventricular septum flattens during diastole, which is a sign of volume overload in the right ventricle. Further distension of the right ventricle will push the interventricular septum towards the left ventricle and reduce the size of the left ventricular chamber. This septal displacement impairs left ventricular filling and increases the left ventricular end-diastolic pressure. In this way, *right-sided heart failure can produce diastolic dysfunction in the left ventricle* by the process known as *interventricular interdependence*.

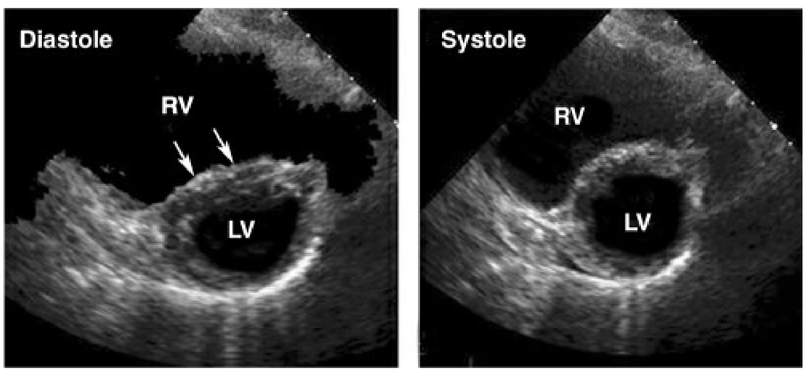

FIGURE 18.2 Transthoracic ultrasound image in the parasternal short-axis view showing right ventricular (RV) enlargement, where the RV cavity is larger than the left ventricular (LV) cavity. Note also that the interventricular septum is flattened during diastole (indicated by the arrows), which is a sign of RV volume overload. Image retouched from Reference 9.

Quantitative measurements of RV chamber size and fractional shortening are also available for the diagnosis of RV failure (9), but measurements obtained with 2D echocardiography are prone to errors because of alterations in the normal RV geometry, and interobserver differences in probe positioning. 3D echocardiography overcomes the problem with altered RV geometry, and provides measures of RVEDV and RVEF (normal RVEF >45%) (10). However, the most accurate measures of RV size and systolic function are provided by cardiac magnetic resonance imaging (not covered here).

Cardiac Filling Pressures

When a pulmonary artery catheter is in place, right heart failure is a consideration if the CVP is equal to, or greater than, the pulmonary artery wedge pressure (PAWP) (11). However, equalization of the CVP and PAWP also occurs with pericardial tamponade, so echocardiography is needed to identify which condition is present.

CARDIOVASCULAR CONSEQUENCES

The physiological and clinical consequences of heart failure are the result of venous congestion and a reduction in forward flow. The extent to which these derangements occur depends on the stage of heart failure, as described next.

Progressive Stages of Heart Failure

The changes in cardiac performance that occur in progressive stages of left heart failure are shown in Figure 18.3. Three distinct stages are identified, and each is summarized below (using the corresponding numbers in Figure 18.3).

. The earliest sign of ventricular dysfunction is an increase in end-diastolic filling pressure (i.e., the pulmonary artery wedge pressure). The stroke volume is maintained, but at the expense of the elevated filling pressure, which produces venous congestion in the lungs and the resulting sensation of dyspnea.

. The next stage is marked by a decrease in stroke volume and an increase in heart rate. The tachycardia offsets the reduction in stroke volume, so the cardiac output is preserved.

. The final stage is characterized by a decrease in cardiac output and a further increase in the ventricular filling pressure. The point at which the cardiac output begins to fall marks the transition from compensated to decompensated heart failure.

Neurohumoral Responses

Heart failure triggers a multitude of endogenous responses, some beneficial and some counterproductive. The following responses have the most clinical relevance (12).

Natriuretic Peptides

Increases in atrial and ventricular wall tension are accompanied by the release of four structurally similar *natriuretic peptides* from cardiac myocytes. These peptides “unload” the ventricles by promoting sodium excretion in the urine (which reduces ventricular preload) and dilating systemic blood vessels (which reduces ventricular preload and afterload). Natriuretic peptides are also used as diagnostic markers for heart failure, as described later.

Sympathetic Nervous System

Decreases in stroke volume are sensed by baroreceptors in the carotid and pulmonary arteries, and activation of these receptors results in brainstem activation of the sympathetic nervous system. This occurs in the early stages of heart failure, and the principal results are positive inotropic and chronotropic effects in the heart, peripheral vasoconstriction, and activation of the renin-angiotensin-aldosterone system.

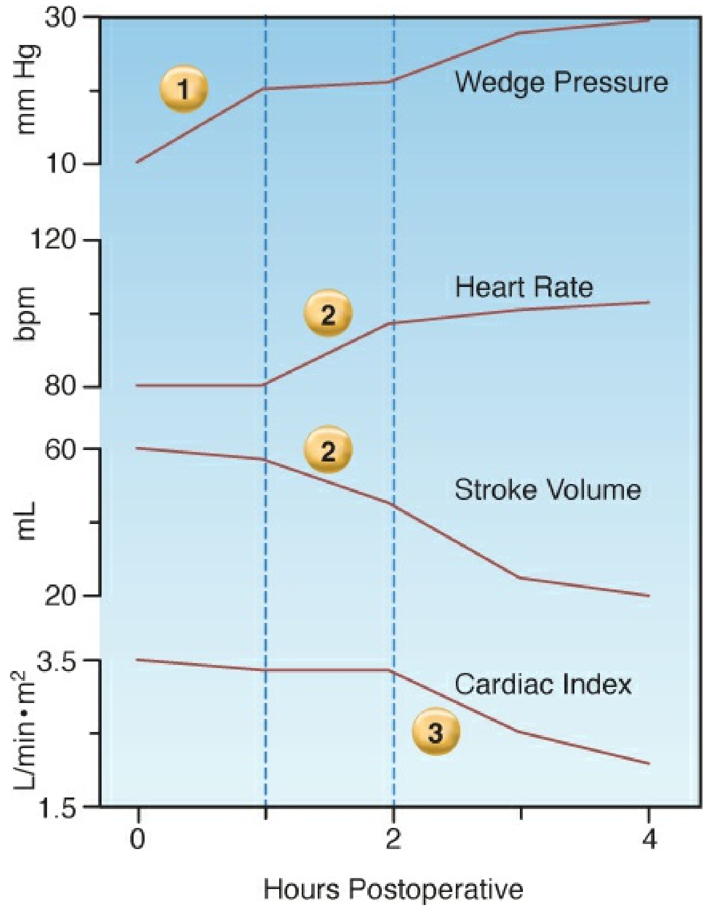

FIGURE 18.3 Changes in cardiac performance during progressive stages of left-sided heart failure in a postoperative patient. Data from personal observations. See text for further explanation.

Renin-Angiotensin-Aldosterone System

Specialized cells in the renal arterioles release renin in response to renal hypoperfusion and adrenergic β-receptor stimulation. Renin release has three consequences: the formation of angiotensin II, the production of aldosterone in the adrenal cortex, and the (angiotensin-triggered) release of arginine vasopressin from the posterior pituitary. Angiotensin produces systemic vasoconstriction, while aldosterone promotes renal sodium and water retention, and vasopressin promotes both vasoconstriction and renal water retention. Angiotensin also helps to preserve the glomerular filtration in the kidneys by vasoconstricting the arterioles on the efferent side of the glomerulus, which increases the filtration pressure across the glomerulus.

Activation of the renin-angiotensin-aldosterone (RAA) system can be counterproductive in the advanced stages of heart failure, when the vasoconstriction can impede systemic blood flow, and the renal retention of sodium and water can promote unwanted (especially pulmonary) edema.

Venous Congestion

Although decompensated heart failure is associated with a decrease in cardiac output, peripheral vasoconstriction is usually sufficient to maintain blood pressure and tissue perfusion. As a result, the principal manifestations of acute, decompensated heart failure are related to venous congestion (13). The following are two of the more serious consequences of venous congestion.

Pulmonary Edema

The clinical presentation of acute heart failure is dominated by complaints of dyspnea and evidence of pulmonary congestion (i.e., basilar crackles on lung auscultation, chest x-ray changes, and hypoxemia). The chest x-ray may show minimal changes, or it may show evidence of pulmonary edema. The chest film in Figure 18.4 shows the classic appearance of “cardiogenic” pulmonary edema, with infiltrates spreading out from the hilar regions bilaterally. Pleural effusions are also commonplace with cardiogenic pulmonary edema. (For a comparison with “noncardiogenic pulmonary edema”, see Chapter 24).

Cardiorenal Syndrome

Heart failure is often accompanied by renal insufficiency, which had been attributed to a reduction in cardiac output. However, there is no correlation between cardiac output and renal dysfunction in patients with heart failure (14), and renal autoregulation typically maintains renal blood flow until the mean arterial pressure drops below 70 mm Hg (15). Observations like these have led to the notion that *the renal dysfunction in acute heart failure is primarily the result of increased venous pressure* (which impairs glomerular blood flow by reducing the renal arteriovenous pressure gradient), and that a low cardiac output plays a role only in patients with hypotension (i.e., cardiogenic shock). This new entity (where venous congestion links heart failure and renal failure) is known as the *cardio-renal syndrome* (5,16), and the principal treatment is diuresis.

Biomarkers

Plasma levels of natriuretic peptides are recommended in patients who present with signs of acute heart failure (2). The principal peptide that is assayed is brain-type or B-type natriuretic peptide (BNP), which is released as a prohormone (proBNP) from both ventricles in response to increased wall tension. The prohormone is cleaved to form BNP (the active hormone) and N-terminal (NT)-proBNP, which is metabolically inactive. The clearance of BNP and NT-proBNP is primarily via the kidneys. However, peptide receptors in adipose tissue also contribute to peptide clearance (17), which may explain why plasma BNP and NT-proBNP levels are lower in obese patients (2). NT-proBNP has a longer half-life than BNP, resulting in plasma levels that are 3–5 times higher than BNP levels.

FIGURE 18.4 Portable chest x-ray in a patient with acute heart failure and pulmonary edema. Note that the infiltrates radiate out from the hilar regions, which is a feature that distinguishes cardiogenic from noncardiogenic pulmonary edema.

TABLE 18.2 Causes of Elevated Natriuretic Peptide Levels

Cardiac	Noncardiac
Heart failure (R and L)	Advanced Age
Myocarditis	Renal Failure
Ventricular Hypertrophy	Anemia
Acute Coronary Syndromes	Pulmonary Embolism
Pericardial Disease	Pulmonary Hypertension
Atrial Fibrillation	Bacterial Sepsis
Cardioversion	Critical Illness
Cardiac Surgery	

From Reference 2.

Diagnostic Value

Heart failure is unlikely if BNP levels are <100 picograms/mL (18,19), or NT-proBNP levels are <300 picograms/mL (20). However, there are several conditions other than heart failure that can elevate natriuretic peptide levels, as shown in Table 18.2. Because of the nonspecific nature of elevated peptide levels (especially in critically ill patients), *natriuretic peptide levels are better suited for excluding the presence of heart failure* (2).

MANAGEMENT STRATEGIES

The management described here is for acute left-sided heart failure that does not result in cardiogenic shock; i.e., is not associated with hypotension or elevated plasma lactate levels. (The management of cardiogenic shock is described in Chapter 16.) As mentioned earlier, this condition typically presents with signs and symptoms related to pulmonary venous congestion. The initial approach is dictated by the blood pressure.

High Blood Pressure

Hypertension is common in acute heart failure (2), and is often a secondary phenomenon (e.g., from agitation, hypoxemia, etc.), but occasionally is the primary problem (e.g., hypertensive emergency). In this situation, management begins with infusion of a vasodilator like the ones in Table 18.3 (2,21).

Vasodilators

The vasodilators in Table 18.3 are capable of dilating both arteries and veins, and will decrease both ventricular preload and afterload. The decrease in preload reduces venous congestion in the lungs, and the decrease in afterload promotes cardiac output. The overall effect is a decrease in arterial blood pressure, an increase in cardiac output, and a decrease in hydrostatic pressure in the pulmonary capillaries.

TABLE 18.3 Vasodilator Infusions for Acute Heart Failure

Agent	Dosing Recommendations
Nitroglycerin	1. Start infusion at 5 μg/min, and increase by 5 μg/min every 5 min to achieve the desired effect. The effective dose is 5–100 μg/min, in most cases, and doses above 200 μg/min are not advised. 2. Tachyphylaxis is common after 24 hrs, and propylene glycol toxicity is a risk with prolonged infusions. 3. Very high doses of nitroglycerin can promote methemoglobinemia.
Nitroprusside	1. Start infusion at 0.2 μg/kg/min and titrate upward every 5 min to achieve the desired effect. The effective dose is 2–5 μg/kg/min in most cases, and the maximum dose allowed is 10 μg/kg/min. 2. To reduce the risk of cyanide toxicity, avoid prolonged infusions at >3 μg/kg/min, and do not use the drug in patients with renal or hepatic insufficiency. Thiosulfate (500 mg) can be added to the infusate to bind the cyanide released from nitroprusside. 3. Not recommended for heart failure associated with coronary ischemia (see text for explanation).

NITROGLYCERIN: Nitroglycerin is the vasodilator of choice for hypertensive acute heart failure (2). The drug binds to soft plastics like polyvinylchloride (PVC), which is a common constituent of the plastic bags and tubing used for intravenous infusions. *As much as 80% of nitroglycerin can be lost to adsorption in PVC-based infusion systems* (22), so it is imperative to use glass bottles and polyethylene tubing when infusing nitroglycerin.

Infusions of nitroglycerin should not continue for longer than 24 hours, because *tachyphylaxis* is common after 24 hours of continuous infusion (2). Prolonged infusions of nitroglycerin can also produce *propylene glycol toxicity* (23), which is characterized by lactic acidosis, delirium, hypotension, and multiorgan failure in severe cases (24), and is a result of the propylene glycol used as a solvent to promote the water solubility of intravenous nitroglycerin. Finally, very high doses of nitroglycerin can produce *methemoglobinemia*, due to the breakdown of nitroglycerin to nitrite ions, which can oxidize hemoglobin to form methemoglobin (25). However, this almost never occurs during therapeutic drug dosing.

NITROPRUSSIDE: About 20% of patients with acute heart failure are resistant to nitroglycerin (2), and in this situation, nitroprusside is available as an alternate vasodilator. Unfortunately, the nitroprusside molecule contains 5 cyanide atoms, and release of the cyanide during nitroprusside breakdown can lead to life-threatening *cyanide intoxication* (26). Both the liver and kidneys participate in cyanide clearance, and thus *nitroprusside is not recommended in patients with renal or hepatic insufficiency*. Thiosulfate binds cyanide and reduces the risk of cyanide toxicity (26), and sodium thiosulfate can be added to nitroprusside infusions as a preventive measure (see Table 18.3).

Nitroprusside has an additional risk in patients with ischemic heart disease because it can produce a *coronary steal syndrome* by diverting blood flow away from ischemic regions of the myocardium (27). Because of this risk, *nitroprusside is not recommended in patients with ischemic heart disease.*

Diuretics

Acute treatment with an intravenous diuretic (described later) is considered only after vasodilator therapy has normalized the blood pressure, *and* there is continued evidence of venous congestion. The most popular diuretic, *intravenous furosemide, produces an acute vasoconstrictor response* (28) by stimulating renin release and promoting the formation of angiotensin II, a potent vasoconstrictor. Because of this response, intravenous furosemide should delayed until the blood pressure is controlled with vasodilator therapy.

For patients with indwelling pulmonary artery catheters, the desired pulmonary artery wedge pressure in left heart failure is the highest pressure that will augment cardiac output without producing pulmonary edema. This pressure usually corresponds to a wedge pressure of 18 to 20 mm Hg (29). Therefore, diuretic therapy can be added if the wedge pressure remains above 20 mm Hg during vasodilator therapy. The specifics of diuretic therapy are described later.

Normal Blood Pressure

For acute heart failure with a normal blood pressure, vasodilator therapy with nitroglycerin is a consideration if there is evidence of reduced peripheral blood flow (e.g., increased PCO_2 gap). Otherwise, an intravenous diuretic (most often furosemide) is indicated, and there is evidence that *administration of a diuretic within one hour of presentation is associated with improved outcomes* (30).

Diuretic Therapy

At the outset, there are three concerns about intravenous diuretics in acute heart failure that deserve mention:

. Intravenous furosemide causes a *decrease* in cardiac output in acute heart failure (31–33). This is the combined result of a decrease in venous return and an increase in left ventricular afterload; the latter effect being attributed to an acute vasoconstrictor response mentioned earlier (28).

- The presence of pulmonary edema in acute heart failure is not evidence of fluid overload, and could be the result of an increase in pulmonary venous pressures from ischemic myocardial "stunning" (often called "flash" pulmonary edema). Diuretics should be used cautiously in this situation.
- Diuretics should be used cautiously in acute heart failure with preserved ejection fraction (HFpEF) because there can be a decrease in cardiac filling in this form of heart failure.

Evidence of a decrease in tissue perfusion after an intravenous diuretic should prompt consideration of vasodilator therapy (e.g., with nitroglycerin) (33), and sometimes a bolus dose of intravenous fluids is advantageous in this situation.

TABLE 18.4 Diuretic Therapy for Acute Heart Failure

Agent	Dosing Recommendations
Furosemide	1. For patients who are furosemide-naïve, start with an IV dose of 40 mg (for normal renal function) or 60–80 mg (for renal impairment). 2. For patients already receiving furosemide, start with an IV dose equal to the total daily furosemide dose. 3. If response not satisfactory after 2 hrs, double the dose, and continue this, if necessary, to a maximum dose of 200 mg. 4. The effective IV dose is then given twice daily.
Bumetanide Torsemide	1. These loop diuretics have a greater bioavailability than furosemide, and can be effective in cases of furosemide resistance. 2. Dose equivalence: 40 mg furosemide = 1 mg bumetanide = 20 mg torsemide
Metolazone	1. A thiazide diuretic that can augment the effect of loop diuretics. 2. Dose is 2.5–10 mg PO daily, and should be given a few hours prior to the loop diuretic. 3. Combination therapy increases the risk of hypokalemia.

FUROSEMIDE: Furosemide is the most popular diuretic for the treatment of heart failure, and acts by blocking sodium reabsorption in the ascending loop of Henle (a "loop diuretic"). The following are some relevant points about furosemide dosing in acute heart failure. (See also Table 18.4.)

- Following an intravenous dose of furosemide, diuresis begins within 5 minutes, peaks at 20–60 minutes, and lasts 2 hours (34).
- For patients who are not on outpatient therapy with furosemide, the initial intravenous dose is 40 mg if renal function is normal. Higher doses (60–80 mg) are needed in the presence of renal insufficiency (35).
- For patients who are receiving furosemide as an outpatient, the initial IV dose should be equivalent to the total daily outpatient dose (35).

. If the diuresis is not adequate (at least 1 liter) at 2 hours after the initial dose, the dose is doubled, and this is continued until a satisfactory response is achieved, or the furosemide dose reaches 200 mg. Failure to respond to an IV dose of 200 mg is evidence of furosemide resistance.

. The effective IV dose of furosemide is then given twice daily. When the venous congestion (e.g., pulmonary edema) is no longer problematic, the drug can be given orally. Although IV and oral doses are often used interchangeably, only 50% of an oral dose of furosemide is absorbed (36), so oral dosing may need to be adjusted upwards to maintain an equivalent response.

Furosemide Resistance

Critically ill patients can have an attenuated response to furosemide, particularly with continued use. This may be the result of hypoalbuminemia (because furosemide is highly protein-bound), renal hypoperfusion, or “diuretic braking’ (i.e., decreased responsiveness as hypervolemia resolves) (37). The following options are available when the diuretic response to furosemide is inadequate.

TRY ANOTHER LOOP DIURETIC: Bumetanide and torsemide are loop diuretics that have a greater bioavailability than furosemide, and will occasionally produce satisfactory diuresis in cases of furosemide resistance. For equivalent dosing, *40 mg of furosemide is equivalent to 1 mg of bumetanide or 20 mg of torsemide* (35).

ADD A THIAZIDE: Thiazide diuretics have a different mechanism of action than loop diuretics (i.e., they block sodium reabsorption in the distal renal tubules), and addition of a thiazide can augment the diuretic response to furosemide. The thiazide most favored for this purpose is *metolazone* because it retains its efficacy in renal insufficiency (37). The dose of metolazone is 2.5–10 mg daily in a single oral dose. The response to metolazone begins at one hour and peaks at 9 hours, so the metolazone dose should be given a few hours prior to the furosemide dose. One concern that deserves mention is that *combined treatment with thiazides and loop diuretics increases the risk of hypokalemia*.

Continuous-Infusion Diuretics

For patients who are massively volume-overloaded, continuous infusions of a loop diuretic can produce a more sustained diuresis than bolus drug injection (38), although this is not a consistent finding (39). The following is a dosing regimen for continuous-infusion furosemide (40):

. Start with an IV bolus dose of 80 mg.

. If the estimated GFR is ≥30 mL/1.73 m^2, start the furosemide infusion at 5 mg/hr. If this does not produce the desired response (i.e., urine output ≥100 mL/hr), give a second bolus dose and increase the infusion rate to 10 mg/hr. Thereafter, the infusion rate can be increased, if needed, to a maximum rate of 40 mg/hr.

. If the estimated GFR is <30 mL/1.73 m^2, start the furosemide infusion at 20 mg/hr. If this does not produce the desired response (i.e., urine output ≥100 mL/hr), give a second bolus dose and increase the infusion rate to 40 mg/hr.

Long-Term Therapies

The following therapies, which are outlined in Table 18.5, have shown evidence of improved long-term outcomes and reduced hospitalizations for patients with heart failure (2). *The principal benefit of these therapies is in heart failure associated with reduced ejection fraction* (HFrEF) (2), but they are also popular in cases of heart failure with preserved ejection fraction (HFpEF), if tolerated. All treatments are oral, and are started (or continued) after the acute episode has resolved, and the patient's condition has stabilized. These therapies are NOT advised for patients with hypotension or evidence of impaired tissue perfusion.

TABLE 18.5 Therapies that Improve Outcomes in Heart Failure with Reduced Ejection Fraction

Agents	Initial Dose	Target Dose
RAA Suppression: Sacubitril/Valsartan	24 mg/26 mg BID for normotension 49 mg/51 mg BID for hypertension	97 mg/103 mg BID
Beta Blockers: Carvedilol Metoprolol succinate extended release	3.125 mg BID 12.5–25 mg once daily	25–50 mg BID 200 mg once daily
SGLT2 Inhibitors: Dapagliflozin or Empagliflozin	10 mg once daily	10 mg once daily
Aldosterone Antagonists: Spironolactone Eplerenone	12.5–25 mg once daily 25 mg once daily	25 mg once daily 50 mg once daily

From the clinical practice guidelines in Reference 2.

RAA Suppression

Drugs that inhibit the renin-angiotensin-aldosterone (RAA) system are recommended for all patients with HFrEF (2). RAA suppressant drugs include angiotensin converting enzyme (ACE) inhibitors, angiotensin-receptor blockers (ARBs), and the combination of an ARB and a neprilysin inhibitor. (Neprilysin is an enzyme that degrades natriuretic peptides.)

All RAA suppressant drugs are considered effective in improving long-term outcomes in patients with HFrEF, and may derive their benefit via left ventricular remodeling (2). *The most effective drug (at this time) is a combination drug that contains sacubitril (a neprilysin inhibitor) and valsartan (an ARB)* (2,41). The dosing regimen is shown in Table 18.5. Low doses are given initially (especially in patients who do not have hypertension), and the dose is gradually increased to the target level over a 4–8 week period (41).

CONCERNS: There are several potential complications with RAA suppression, including hypotension, worsening renal function, hyperkalemia, and angioedema. These drugs are contraindicated in patients with a history of angioedema from any cause. Also worthy of mention is that RAA suppression has shown no consistent benefit in patients with heart failure associated with preserved ejection fraction (HFpEF) (2).

Beta Blockers

Long-term treatment with beta blockers is recommended for all patients with HFrEF (2). The drugs that have proven effective in improving long-term outcomes include *carvedilol* and *sustained-release metoprolol (succinate)*, and the dosing regimens for these agents is shown in Table 18.5. *De novo* therapy is started at low doses and gradually advanced to target levels.

CONCERNS: Sudden withdrawal of beta blockers can result in a hyperadrenergic state known as "beta blocker rebound" (42). Therefore, for patients on long-term beta blocker therapy, treatment should be restarted as soon as the patient's condition has stabilized. Also worthy of mention is that the current guidelines for managing heart failure (2) includes no recommendation for beta blockers in heart failure associated with preserved ejection fraction.

SGLT2 Inhibitors

For unclear reasons, oral hypoglycemic therapy with sodium-glucose cotransporter 2 (SGLT2) inhibitors is associated with improved long-term outcomes in heart failure. This effect is independent of the glucose-lowering effect of the drugs, and occurs in both diabetic and nondiabetic patients (43). SGLT2 inhibitors are recommended for all patients with advanced heart failure (both HFrEF and HFpEF), regardless of the presence or absence of diabetes (2). The specific drugs and recommended doses are shown in Table 18.5.

CONCERNS: There is an increased risk of *euglycemic diabetic ketoacidosis* with SGLT2 inhibitors (44).

Aldosterone Antagonists

Aldosterone antagonists block sodium reabsorption in the renal collecting ducts, and can augment the diuretic effect of loop diuretics. These drugs (i.e., spironolactone, eplerenone) have shown a survival benefit in patients with HFrEF (2), and the recommended dosing regimens are shown in Table 18.5. As with all diuretics, these drugs should be used cautiously in cases of heart failure associated with a preserved ejection fraction, where the diastolic dysfunction can impair ventricular filling.

CONCERNS: Aldosterone antagonists can promote hyperkalemia, and should not be used in patients with renal failure (i.e., GFR <30 mL/min/1.73 m^2) or recent problems with hyperkalemia.

Positive Pressure Breathing

Positive intrathoracic pressures will reduce left ventricular afterload by decreasing the transmural wall pressure developed by the ventricle during systole. This promotes ventricular emptying by facilitating the inward movement of the ventricular wall during systole. As a result, positive pressure breathing can augment the stroke output of the left ventricle (see Figure 11.5).

Clinical studies have demonstrated that breathing with continuous positive airway pressure (CPAP) increases cardiac output in patients with left-sided heart failure (45), and hastens clinical improvement in patients with cardiogenic pulmonary edema (46,47). As a result of these observations, noninvasive positive-pressure breathing has become an accepted treatment modality for acute heart failure associated with pulmonary edema.

A FINAL WORD

Be Cautious with Diuretics

The principle consequence in heart failure (other than cardiogenic shock) is venous congestion, hence the popular term “congestive heart failure”, and the popular use of diuretics. However, the following are reasons to avoid the overzealous use of diuretic therapy in patients with acute decompensated heart failure.

- The presence of acute, cardiogenic pulmonary edema is not always evidence of fluid overload, as the problem may be acute myocardial stiffening from cardiac ischemia, as seen in “flash pulmonary edema”.
- Intravenous furosemide causes a decrease in cardiac output in acute heart failure (31–33) via the dual effect of a drug-induced increase in left ventricular afterload (28) and a diuresis-induced decrease in ventricular preload.
- The tendency for diuresis to decrease cardiac output will be magnified in the 50% of cases of heart failure due to myocardial stiffening (i.e., heart failure with preserved ejection fraction), which impairs cardiac filling during diastole.

The above concerns might explain the observation that diuretic management in acute heart failure improves outcomes only when the diuretic dosing is limited (48).

References

1. Tsao CW, Aday AW, Almarzoog ZI, et al. Heart disease and stroke statistics—2023 update: A report from the American Heart Association. Circulation 2023; 147:e93–e621.

Guidelines and Reviews

2. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure. A report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. J Am Coll Cardiol 2022; 79:e263–e421.
3. McDonagh TA, Metra M, Asamo M, et al; ESC Scientific Document Group. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. Eur Heart J 2021; 42:3599–3726.
4. Kanwar MK, Everett KD, Gulati G, et al. Epidemiology and management of right ventricular-predominant heart failure and shock in the cardiac intensive care unit. Eur Heart J Acute Cardiovasc Care 2022; 11:584–594.
5. Rangaswami J, Bhalla V, Blair JEA, et al. Cardiorenal syndrome: classification, pathophysiology, diagnosis and treatment strategies. A scientific statement from the American Heart Association. Circulation 2019; 139:e840–e878.

Types of Heart Failure

6. Paulus WJ, Tschope C, Sanderson JE, et al. How to diagnose diastolic heart failure: a consensus statement on the diagnosis of heart failure with normal left ventricular ejection fraction by the Heart Failure and Echocardiography Associations of the European Society of Cardiology. Eur Heart J 2007; 28:2539–2550.
7. Hurford WE, Zapol WM. The right ventricle and critical illness: a review of anatomy, physiology, and clinical evaluation of its function. Intensive Care Med 1988; 14:448–457.
8. Prada G, Pustavoitau A, Koenig S, et al. Focused cardiac ultrasonography for right ventricular size and function. N Engl J Med 2022; 387:e52.
9. Rudski LG, Lai WW, Afilalo J, et al. Guidelines for the echocardiographic assessment of the right heart in adults: a report from the American Society of Echocardiography, endorsed by the European Association of Echocardiography, a registered branch of the European Society of Cardiology, and the Canadian Society of Echocardiography. J Am Soc Echocardiogr 2010;

23:685–713.
10. Partington SL, Kilner PJ. How to image the dilated right ventricle. Circulation: Cardiovascular Imaging. 2017; 10:e004688.
11. Lopez-Sendon J, Coma-Canella I, Gamello C. Sensitivity and specificity of hemodynamic criteria in the diagnosis of right ventricular infarction. Circulation 1981; 64:515–525.

Cardiovascular Consequences

12. Mann DL. Pathophysiology of heart failure. In: Libby P, Bonow RO, Mann DL, Zipes DP, eds. Braunwald's Heart Disease. 8th ed. Philadelphia: Saunders Elsevier, 2008;541–560.
13. Gheorghiade M, Pang PS. Acute heart failure syndromes. J Am Coll Cardiol 2009; 53:557–573.
14. Hanberg JS, Sury K, Wilson FP, et al. Reduced cardiac index is not the dominant driver of renal dysfunction in heart failure. J Am Coll Cardiol 2016; 17:2199–2208.
15. Burke M, Pabbidi MR, Farley J, Roman RJ. Molecular mechanisms of renal blood flow autoregulation. Curr Vasc Pharmacol 2014; 12:845–858.
16. Thind GS, Loehrke M, Wilt JL. Acute cardiorenal syndrome: Mechanisms and clinical implications. Clev Clin J Med 2018; 85:231–239.
17. Wang TJ. The natriuretic peptides and fat metabolism. N Engl J Med 2012; 367:377–378.
18. Maisel AS, Krishnaswamy P, Nomak RM, et al. Rapid measurement of B-type natriuretic peptide in the emergency diagnosis of heart failure. N Engl J Med 2002; 347:161–167.
19. Maisel AS, McCord J, Nowak J, et al. Bedside B-type natriuretic peptide in the emergency diagnosis of heart failure with reduced or preserved ejection fraction. Results from the Breathing Not Properly Multinational Study. J Am Coll Cardiol 2003; 41:2010–2017.
20. Januzzi JL, van Kimmenade R, Lainchbury J, et al. NT-proBNP testing for diagnosis and short-term prognosis in acute destabilized heart failure: an international pooled analysis of 1256 patients. Eur Heart J 2006; 27:330–337.

Management Strategies

21. Rhoney D, Peacock WF. Intravenous therapy for hypertensive emergencies, part 1. Am J Health Syst Pharm 2009; 66:1343–1352.
22. Nitroglycerin. In: AHFS Drug Information, 2022. Bethesda: American Society of Health System Pharmacists, 2022;1755–1759.
23. Demey HE, Daelemans RA, Verpooten GA, et al. Propylene glycol-induced side effects during intravenous nitroglycerin therapy. Intensive Care Med 1988; 14:221–226.
24. Zar T, Graeber C, Perazella MA. Recognition, treatment, and prevention of propylene glycol toxicity. Semin Dial 2007; 20:217–219.
25. Curry SC, Arnold Cappell P. Nitroprusside, nitroglycerin, and angiotensin- converting enzyme inhibitors. Crit Care Clin 1991; 7:555–582.
26. Hall VA, Guest JM. Sodium nitroprusside-induced cyanide intoxication and prevention with sodium thiosulfate prophylaxis. Am J Crit Care 1992; 1:19–25.
27. Mann T, Cohn PF, Holman LB, et al. Effect of nitroprusside on regional myocardial blood flow in coronary artery disease. Results in 25 patients and comparison with nitroglycerin. Circulation 1978; 57:732–738.
28. Francis GS, Siegel RM, Goldsmith SR, et al. Acute vasoconstrictor response to intravenous furosemide in patients with chronic congestive heart failure. Ann Intern Med 1986; 103:1–6.
29. Franciosa JA. Optimal left heart filling pressure during nitroprusside infusion for congestive heart failure. Am J Med 1983; 74:457–464.
30. Matsue Y, Damman K, Voors AA, et al. Time-to-furosemide treatment and mortality in patients hospitalized with acute heart failure. J Am Coll Cardiol 2017; 69:3042–3051.
31. Kiely J, Kelly DT, Taylor DR, Pitt B. The role of furosemide in the treatment of left ventricular dysfunction associated with acute myocardial infarction. Circulation 1973; 58:581–587.
32. Mond H, Hunt D, Sloman G. Haemodynamic effects of frusemide in patients suspected of having acute myocardial infarction. Br Heart J 1974; 36:44–53.
33. Nelson GIC, Ahuja RC, Silke B, et al. Haemodynamic advantages of isosorbide dinitrate over frusemide in acute heart failure following myocardial infarction. Lancet 1983a; i:730–733.
34. Furosemide monograph. In AHFS Drug Information, 2022. Bethesda, MD: American Society of Health System Pharmacists, 2022;2645–2650.
35. Colucci WS. Treatment of acute decompensated heart failure: Specific therapies. UpToDate (www.uptodate.com). Accessed October 12, 2023.
36. Brater DC. Diuretic therapy N Engl J Med 1998; 339:387–395.
37. Asare K, Lindsey K. Management of loop diuretic resistance in the intensive care unit. Am J Health Syst Pharm 2009; 66:1635–1640.
38. Salvador DR, Rey NR, Ramos GC, Punzalan FE. Continuous infusion versus bolus injection of loop diuretics in congestive heart failure. Cochrane Database Syst Rev 2005;CD003178.

39. Felker GH, Lee KL, Bull DA, et al. Diuretic strategies in acute heart failure. N Engl J Med 2011; 364:797–805.
40. Brater DC, Ellison DH. Causes and treatment of refractory edema in adults. UpToDate (www.uptodate.com). Accessed October 14, 2023.
41. Velazquez EJ, Morrow DA, DeVore AD, et al. Angiotensin-neprilysin inhibition in acute decompensated heart failure. N Engl J Med 2019; 380:539–548.
42. Koracevic G, Micic S, Stojanovic M. By discontinuing beta blockers before an exercise test, we may precipitate a rebound phenomenon. Curr Vasc Pharmacol 2021; 19:624–633.
43. Zannad F, Ferreira JP, Pocock SJ, et al. SGLT2 inhibitors in patients with heart failure with reduced ejection fraction: a meta-analysis of the EMPEROR-Reduced and DAPA-HF trials. Lancet 2020; 396:819–829.
44. Rosenstock J, Ferrannini E. Euglycemic diabetic ketoacidosis: A predictable, detectable, and preventable safety concern with SGLT2 inhibitors. Diabetes Care 2015; 38:1638–1642.
45. Bradley TD, Holloway BM, McLaughlin PR, et al. Cardiac output response to continuous positive airway pressure in congestive heart failure. Am Rev Respir Crit Care Med 1992; 145:377–382.
46. Nouira S, Boukef R, Bouida W, et al. Non-invasive pressure support ventilation and CPAP in cardiogenic pulmonary edema: a multicenter randomized study in the emergency department. Intensive Care Med 2011; 37:249–256.
47. Ducros L, Logeart D, Vicaut E, et al. CPAP for acute cardiogenic pulmonary edema from out-of-hospital to cardiac intensive care unit: a randomized multicenter study. Intensive Care Med 2011; 37: 1501–1509.

A Final Word

48. Peacock WF, Costanzo MR, De Marco T, et al. Impact of intravenous loop diuretics on outcomes of patients hospitalized with acute decompensated heart failure: insights from the ADHERE registry. Cardiology 2009; 113:12–19.

Chapter 19

Tachyarrhythmias

Variability is the law of life . . .

William Osler (*a*)

A rapid heart rate or *tachycardia* while at rest is evidence of a problem, but the tachycardia itself may not be the problem. This chapter describes tachycardias that *are* a problem (i.e., *tachyarrhythmias*), and require prompt evaluation and management. Many of the recommendations in this chapter are borrowed from the clinical practice guidelines listed in the bibliography at the end of the chapter (1–6).

RECOGNITION

The diagnostic evaluation of tachycardias (heart rate >100 beats/min) is based on 3 findings on the ECG: i.e., the duration of the QRS complex, the uniformity of the R-R intervals, and the characteristics of the atrial activity. This approach is outlined in the flow diagram in Figure 19.1. The duration of the QRS complex is used to distinguish *narrow-QRS-complex tachycardias* (QRS duration ≤0.12 sec) from *wide-QRS-complex tachycardias* (QRS duration >0.12 sec), which helps to identify the point of origin of the tachycardia, as described next.

Narrow-QRS-Complex Tachycardias

Tachycardias with a narrow QRS complex (≤0.12 sec) originate from a site above the AV conduction system, and are also known as *supraventricular tachycardias*. These include sinus tachycardia, atrial tachycardia, AV nodal re-entrant tachycardia (also called paroxysmal supraventricular tachycardia), atrial flutter, and atrial fibrillation. The specific arrhythmia can be identified using the uniformity of the R-R interval (i.e., the regularity of the rhythm), and the characteristics of the atrial activity, as described next.

Regular Rhythm

If the R-R intervals are uniform in length (indicating a regular rhythm), the possible arrhythmias include sinus tachycardia, AV nodal re-entrant tachycardia, and atrial flutter with a fixed (2:1, 3:1) AV block. The atrial activity on the ECG can identify each of these rhythms using the following criteria:

- Uniform P waves and P–R intervals are evidence of sinus tachycardia.
- The absence of P waves suggests an AV nodal re-entrant tachycardia (see Figure 19.2).
- Sawtooth waves are evidence of atrial flutter.

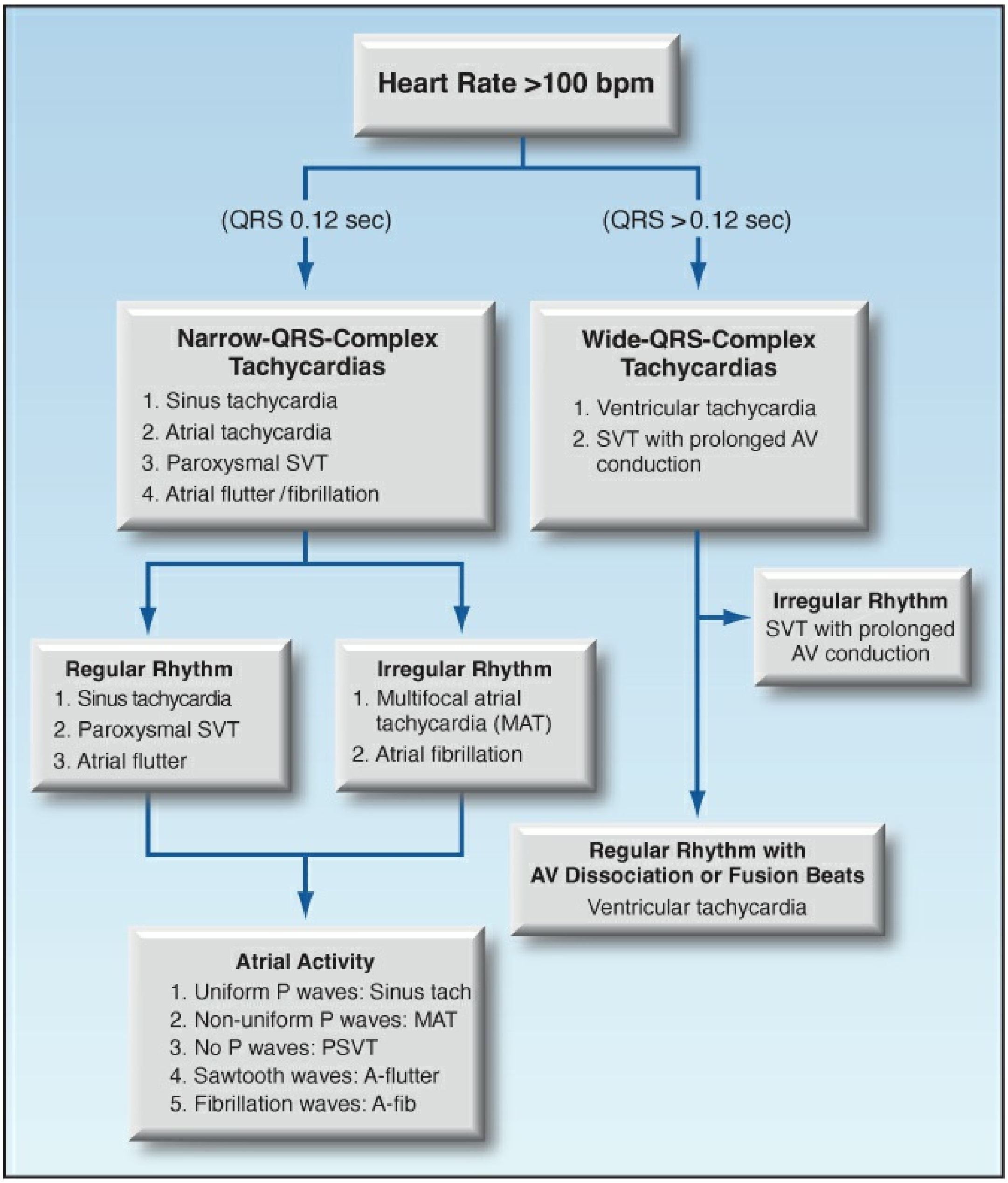

FIGURE 19.1 Flow diagram for the evaluation of tachycardias. See text for further details.

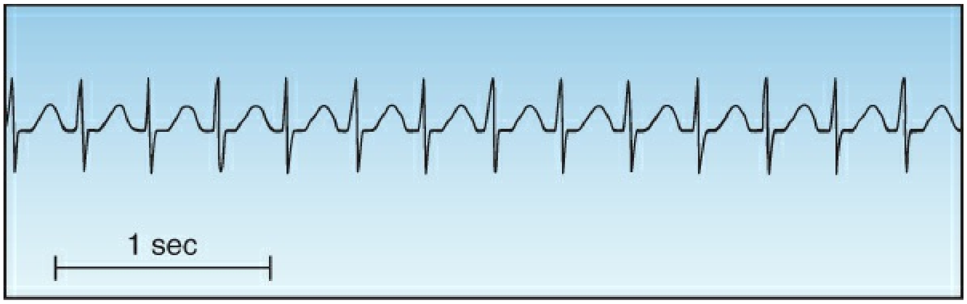

FIGURE 19.2 Narrow-QRS-complex tachycardia with a regular rhythm. Note the absence of visible P waves. This is an AV nodal re-entrant tachycardia, also known as a "paroxysmal supraventricular tachycardia" (paroxysmal SVT) because of its abrupt onset.

Irregular Rhythm

If the R-R intervals are not uniform in length (indicating an irregular rhythm), the most likely arrhythmias are multifocal atrial tachycardia and atrial fibrillation. Once again, the atrial activity on the ECG helps to identify each of these rhythms; i.e.,

. Multiple P wave morphologies and variable PR intervals are evidence of multifocal atrial tachycardia (see Panel A, Figure 19.3).

. The absence of P waves with highly disorganized atrial activity (fibrillation waves) is evidence of atrial fibrillation (see Panel B, Figure 19.3).

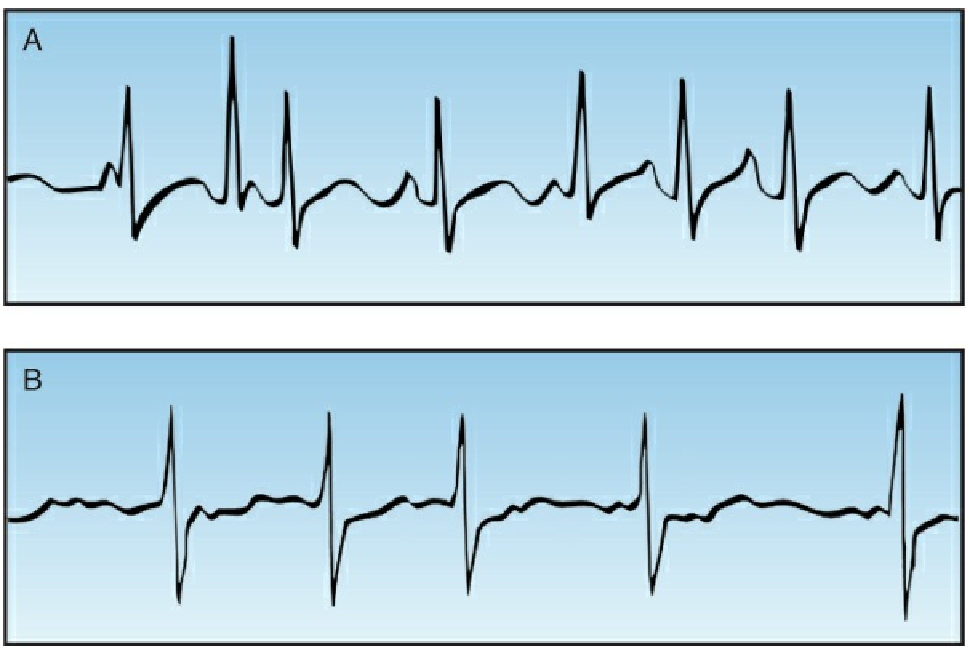

FIGURE 19.3 Narrow-QRS-complex tachycardias with an irregular rhythm. Panel A shows a multifocal atrial tachycardia (MAT), identified by multiple P wave morphologies and variable PR intervals. Panel B is atrial fibrillation, identified by the absence of P waves with highly disorganized atrial activity (fibrillation waves).

Wide- QRS- Complex Tachycardias

Tachycardias with a wide QRS complex (>0.12 sec) can originate from a site below the AV conduction system (i.e., ventricular tachycardia), or they can represent a supraventricular tachycardia (SVT) with prolonged AV conduction (e.g., from a bundle branch block). These two arrhythmias can be difficult to distinguish. An irregular rhythm is evidence of an SVT with aberrant AV conduction, while certain ECG abnormalities (e.g., AV dissociation) provide evidence of VT. The distinction between VT and SVT with aberrant conduction is described in more detail later in the chapter.

ATRIAL FIBRILLATION

Atrial fibrillation (AF) is the most common cardiac arrhythmia in adults (prevalence = 2–4%), and is classified as paroxysmal (present ≤7 days), persistent (present for >7 days), long-standing persistent (present for >1 year), or permanent (when repeated attempts to restore sinus rhythm are unsuccessful) (3). Most patients are elderly (median age = 75 years) and have underlying cardiovascular or pulmonary disease (with the exception of hyperthyroidism). AF can also appear *de novo* in certain groups of hospitalized patients (see next).

Hospital-Related AF

New-onset AF is a reported risk in ICU patients, and in the postoperative period following major surgery.

ICU-Related AF

The reported incidence of ICU-related AF varies widely (from 2% to 44%), and is more likely to appear in elderly patients with coronary disease, sepsis, shock, or respiratory failure (7,8). The AF resolves spontaneously in up to 25% of cases (7), but it tends to recur, and rehospitalization for AF is reported in 20% of cases (8). Stroke is a risk, but the reported incidence is only 1% in the ensuing 3 years (8).

Postoperative AF

Postoperative AF is reported in up to 45% of cases involving cardiac and non-cardiac thoracic surgery, and up to 10% of cases involving other major surgeries (7,9). It usually appears in the first 5 postoperative days (9), and follows the same course as ICU-related AF. Prophylaxis with β-blockers and magnesium has been effective in the AF that follows cardiac surgery AF (10).

Adverse Consequences

AF can impair cardiac performance and increase the risk of thromboembolic stroke.

Cardiac Performance

Atrial contraction is responsible for about 25% of the ventricular end diastolic volume (11). Loss of this atrial contribution (from AF) has little noticeable effect unless diastolic filling is already impaired by mitral stenosis or reduced ventricular compliance (e.g., from ventricular hypertrophy). Under these conditions, a rapid heart rate can decrease cardiac stroke output (because of the decreased time for ventricular filling).

The influence of ventricular filling time in AF is illustrated by the arterial pressure waveforms in Figure 19.4. Note that close spacing between two heart beats (which shortens the time for ventricular filling), results in a decrease in systolic pressure (which is a reflection of the stroke volume). This effect can result in a "pulse deficit" (i.e., a decrease in the peripheral pulse rate relative to the apical impulses), indicating that the stroke output of the heart is not always reaching the periphery. Although rarely mentioned, monitoring the pulse deficit could be useful for identifying the end-point of rate control in AF (12).

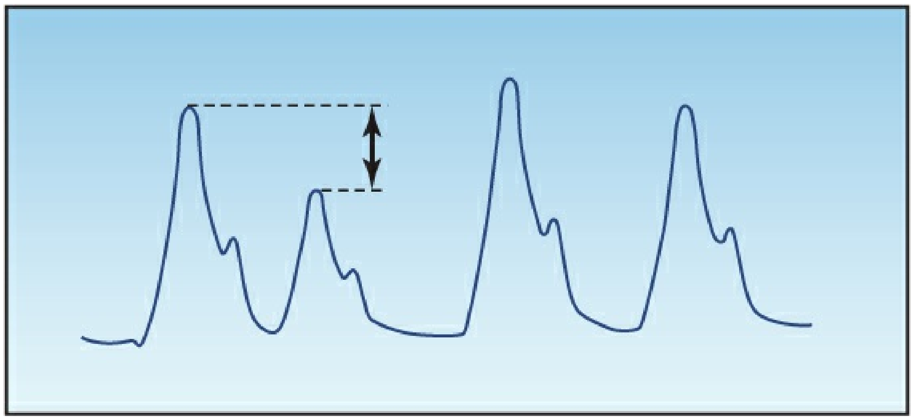

FIGURE 19.4 Arterial pressure tracings in AF. Note that there is a decrease in systolic pressure when two heart beats occur in rapid succession. This effect can produce a "pulse deficit", where apical impulses are not always transmitted to the periphery.

Stroke

Atrial fibrillation predisposes to thrombus formation in the left atrium, especially in the left atrial appendage, and these thrombi can dislodge and embolize in the cerebral circulation to produce an acute ischemic stroke. Overall, AF increases the risk of stroke 5-fold (3–5), but the risk is not homogeneous, and is influenced by the presence of certain risk factors (which are described later). The increased risk of stroke pertains to all types of AF, except first episodes of AF that are <48 hrs in duration. Recommendations for stroke-prevention with antithrombotic therapy are presented later.

Heart Rate Control

In patients with rapid AF who are hemodynamically stable, the initial goal is to slow the heart rate with drugs that prolong AV conduction. A variety of drugs are available for this purpose (13), and the popular ones are included in Table 19.1. The following is a brief description of each drug. (*Caveat:* These drugs should not be used if the AF originates from an accessory pathway in the AV node, as described later).

TABLE 19.1 Drug Regimens for Acute Rate Control in Atrial Fibrillation

Drug	Dosing Regimens and Comments
Diltiazem	Dosing: 0.25 mg/kg IV over 2 min, then infuse at 5–15 mg/hr. Comment: A popular agent for acute rate control, but use is limited by the risk of hypotension (20–30% of cases), and by negative inotropic effects.

Amiodarone	Dosing: 150 mg IV over 10 min, and repeat if needed, then infuse at 1 mg/min for 6 hr and 0.5 mg/min for 18 hr. Total dose should not exceed 2.2 grams in 24 hrs. Comment: An effective alternative to diltiazem and beta blockers, and may be preferred in HFrEF. Can occasionally promote conversion to sinus rhythm.
Metoprolol	Dosing: 2.5–5 mg IV over 2 min, and repeat every 5–10 min if needed to a total of 3 doses. Comment: Most effective in AF associated with hyperadrenergic states. Similar to diltiazem in risk of hypotension and negative inotropic effects.
Esmolol	Dosing: 500 µg/kg IV bolus, then infuse at 50 µg/kg/min. Increase dose in increments of 25 µg/kg/min every 5 min if needed to a maximum rate of 200 µg/kg/min. Comment: An ultra-short-acting β blocker that permits rapid dose titration. Other features similar to metoprolol.
Digoxin	Dosing: 0.25 mg IV every 2 hrs to a total dose of 1.5 mg, then 0.125–0.375 mg IV daily. Comment: Slow-acting drug that should not be used alone for acute rate control. Used primarily for long-term rate control in patients with HFrEF.

See text for references. HFrEF = heart failure with reduced ejection fraction.

Diltiazem

The most popular agent for acute rate control in AF is the calcium channel blocker diltiazem, which is given as an intravenous bolus followed by a continuous infusion. The bolus dose produces effective rate control within one hour in 55% of patients (14), and the continuous infusion maintains rate control in up to 90% of cases (15). The graph in Figure 19.5 shows that diltiazem is more effective than either amiodarone or digoxin in the first few hours of drug therapy (16).

COMPLICATIONS: The use of diltiazem is limited by the risk of hypotension, which is reported in about 20–30% of cases (15,17). Diltiazem also has negative inotropic effects, which limits its use in patients with heart failure associated with reduced ejection fraction.

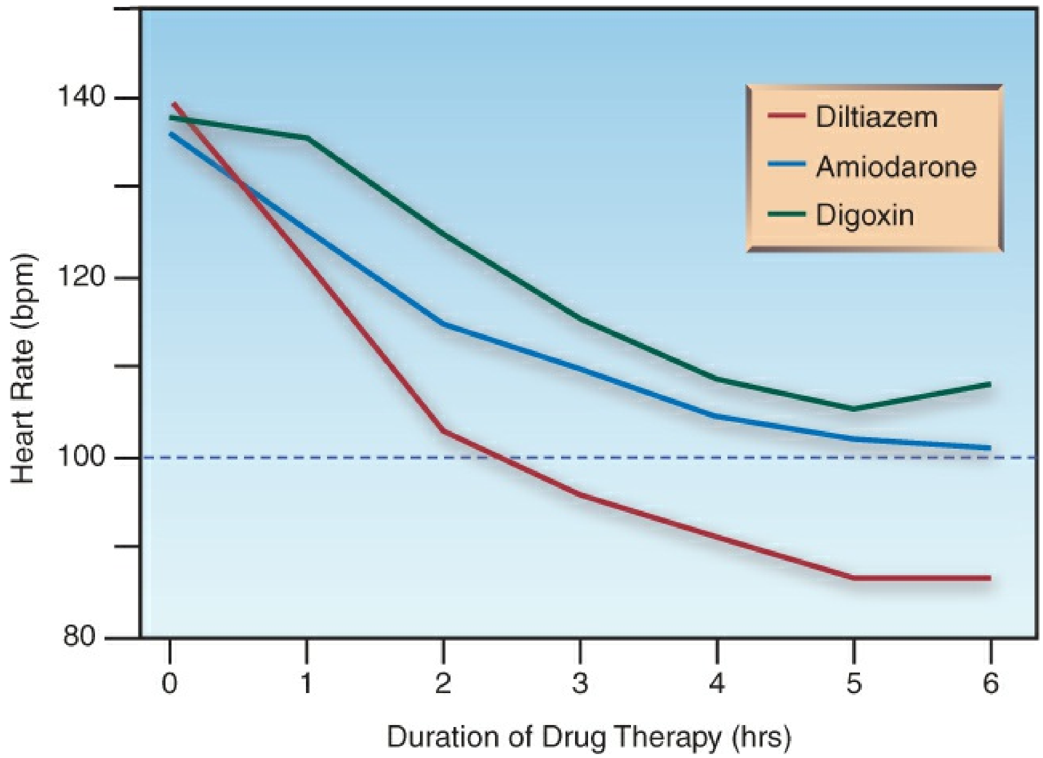

FIGURE 19.5 Comparison of acute rate control with intravenous diltiazem, amiodarone, and digoxin in patients with uncomplicated atrial fibrillation. Data from Reference 16.

β-Receptor Antagonists

β-blockers achieve successful rate control in 70% of cases of rapid AF (18), and are most effective in hyperadrenergic states. Two β-blockers with proven efficacy in acute rate control are *esmolol* (Brevibloc) and *metoprolol* (Lopressor), and their dosing regimens are shown in Table 19.1. Both are cardioselective agents that preferentially block β-1 receptors in the heart, but both are limited by the risk of hypotension. Esmolol is more appealing than metoprolol because it is an ultra short-acting drug (serum half-life is only 9 minutes), which allows rapid dose titration (19).

Amiodarone

Amiodarone is an effective alternative to diltiazem or beta blockers in patients who are resistant or intolerant to these drugs (20). It also produces less cardiac depression than diltiazem or beta blockers, and is favored in some guidelines for patients with heart failure and reduced ejection fraction (4). Amiodarone is also an antiarrhythmic agent (Class III), and is capable of converting AF to a sinus rhythm. However, the success rate for cardioversion is variable, and some studies show no difference in the rate of cardioversion with amiodarone and diltiazem (17).

COMPLICATIONS: Intravenous amiodarone does not have the feared side effects associated with long-term therapy (e.g., pulmonary fibrosis, thyroid dysfunction). Hypotension can occur, but the incidence is far lower than with diltiazem (17). The hypotensive effect can be attributed to a solvent (polysorbate 80 surfactant) used to enhance the water solubility of amiodarone, since the aqueous formulation of amiodarone does not cause hypotension (21). Amiodarone can also prolong the QT interval, and "torsade de pointes" has been reported, although rarely (22). Finally, amiodarone has several drug interactions by virtue of its metabolism by the cytochrome P450 enzyme system in the liver. The most relevant interactions in the ICU setting are inhibition of digoxin and warfarin metabolism.

TRANSITION TO ORAL THERAPY: Transition from IV to oral amiodarone begins with an oral dose of 400 mg, which is given 4–5 hours before discontinuing the infusion. The initial oral regimen is 400 mg twice daily, which is continued until the total loading dose (IV and PO) reaches 10 grams (22). (The initial 24-hour IV regimen is a total dose of about one gram.) Thereafter, the daily dose can be reduced to the usual maintenance dose (100–200 mg once daily).

Digoxin

Digoxin prolongs conduction in the AV node, and is used for long-term rate control in patients with AF and heart failure with reduced ejection fraction. However, its value for acute rate control is limited because *the response to intravenous digoxin is slow to develop*. This is demonstrated in Figure 19.5 (16). Note that there is little effect in the first hour after drug administration, and the heart rate has not reached the desired end-point (<100 bpm) after 6 hours. Because of this delayed response, intravenous digoxin should not be used alone for acute rate control in AF (3,13). It can, however, be used as an adjunct to diltiazem or beta blockers in patients with heart failure or labile BP.

End-Point of Rate Control

There is a surprising lack of agreement about the end-point of rate reduction in AF, and heart rates of 80 bpm to 110 bpm have been used as end-points in individual studies (3). However the phenomenon shown in Figure 19.4 suggests another approach to this issue; i.e., *elimination of the pulse deficit, or resolution of the variable systolic pressure in rapid AF, provides a more physiological goal of heart rate reduction in AF.* This approach is often overlooked, but deserves your consideration.

Electrical Cardioversion

Synchronized, direct-current cardioversion is reserved for hemodynamically unstable patients. Biphasic shocks have a greater success rate and require lower energy levels than monophasic shocks (23). An energy level of 100 joules is usually successful with biphasic shocks (vs. 200 joules for monophasic shocks) (23). Fixed-energy attempts are considered more effective than an escalating energy approach (3). The success rate of cardioversion is greatest for new-onset or recent-onset AF.

Cardiac ultrasound (via the transesophageal approach, if time permits) is required prior to cardioversion to determine the presence or absence of atrial thrombi. Anticoagulation with a direct-acting oral agent is started as soon as possible (prior to, or immediately after the procedure) in appropriate candidates. Anticoagulation is not required for new-onset AF that is less than 48 hours in duration.

Stroke Prevention

As mentioned earlier, there is an increased risk of thromboembolic stroke in AF, but a number of factors influence this risk (as well as the need for anticoagulant prophylaxis).

Risk Assessment

The decision to use anticoagulant therapy in AF is based on a scoring system that is shown in Table 19.2. There are 10 identified risk factors for stroke, with the highest risk being advanced age (>75 years), and a prior history of stroke, transient ischemic attack (TIA) or venous thromboembolism (VTE). Patients who are definite candidates for anticoagulation are considered to have at least a five-fold higher risk of stroke (or a yearly incidence >5%). It is important to mention that the risk assessment in Table 19.2 does not apply to valvular AF (i.e., AF associated with mitral stenosis or a prosthetic valve), which always requires anticoagulation (3).

Contraindications to anticoagulation in AF include active hemorrhage, a history of intracerebral hemorrhage, intracranial tumor, recurrent bleeding from a lesion that is still present, and a platelet count <50,000/µL (3,4). Of interest, *a history of falls is not a contraindication to anticoagulant therapy* (24).

TABLE 19.2 Risk Assessment for Oral Anticoagulation in Atrial Fibrillation

CHA$_2$DS$_2$-VASc Scoring		
Condition	**Points**	**Indications for Anticoagulation**
(C) CHF	1	**Definite** Males: ≥2 points Females: ≥3 points
(H) Hypertension	1	
(A) Age >75 yrs	2	
(D) Diabetes	1	
(S) Stroke/TIA/VTE	2	**Consider** Males: 1 point Females: 2 points
(V) Vascular Disease	1	
(A) Age 65–74 yrs	1	
(Sc) Sex Category (♀)	1	

VTE = venous thromboembolism. From References 3–5.

Oral Anticoagulants

The oral anticoagulants used for stroke prevention in AF are shown in Table 19.3.

Warfarin:

Warfarin is the first oral anticoagulant approved for clinical use (in the 1950s), and acts indirectly by inhibiting the synthesis of vitamin K-dependent clotting factors (Factors II, VII, IX, and X). Anticoagulation with warfarin reduces the incidence of stroke (by as much as 65%) in both valvular and nonvalvular AF (3–5,13,25). However, warfarin has a narrow therapeutic window (i.e., the difference between therapeutic and toxic drug levels), which increases the likelihood of both inadequate and excessive anticoagulation, and requires routine monitoring of hemostasis (with the international normalized ratio, or INR).

TABLE 19.3	Oral Anticoagulants for Atrial Fibrillation
Drug	**Dosing Recomendations**
Dabigatran (Pradaxa)	Standard Dose: 50 mg BID. Reduced Dose: 110 mg BID for age ≥80 yrs or CrCL <30 mL/min.
Rivaroxaban (Xarelto)	Standard Dose: 20 mg once daily. Reduced Dose: 15 mg once daily for CrCL <30 mL/min.
Apixaban (Eliquis)	Standard Dose: 5 mg BID. Reduced Dose: 2.5 mg BID for 2 of the following: age ≥80 yrs, weight ≤60 kg, serum creatinine ≥1.5 mg/dL.
Edoxaban	Standard Dose: 60 mg once daily. Reduced Dose: 30 mg once daily for CrCL <30 mL/min, weight ≤60 kg, or therapy with erythromycin or ketoconazole.
Warfarin (Coumadin)	Starting dose is usually 5 mg once daily, which is adjusted to achieve an INR of 2–3.

From the clinical practice guidelines in Reference 3. CrCL =-creatinine clearance (estimated).

Direct-Acting Oral Anticoagulants

The direct-acting oral anticoagulants (DOACs), which include direct thrombin inhibitors (e.g., dabigatran) and drugs that inhibit Factor Xa (e.g., rivaroxaban, apixaban, and edoxaban), have a more predictable anticoagulant effect than warfarin, and are given in fixed doses, without monitoring any tests of hemostasis. Clinical studies in patients with nonvalvular AF have shown that DOACs are as effective as warfarin in reducing the incidence of stroke, and have a lower incidence of major bleeding (26–29). Based on these observations, *DOACs are preferred to warfarin for stroke reduction in patients with nonvalvular AF* (3–5). Each DOAC is considered equivalent to the others, but edoxaban may be less effective for stroke reduction (29), and dabigatran may have a greater risk of bleeding (26).

The preference for DOACs does not extend to valvular AF (i.e., AF associated with significant mitral stenosis or the presence of any prosthetic valve). In this situation, warfarin is considered superior to DOACs for stroke prevention (3–5).

DOSING: The recommended dosing for each DOAC is shown in Table 19.3. Note that a dose reduction is required for all DOACs in patients with renal insufficiency (creatinine clearance <30 mL/min). In this situation, Factor Xa levels can be measured to assess the adequacy of anticoagulation. The consensus opinion is that DOACs should be avoided in patients with end-stage renal failure (creatinine clearance <15 mL/min) (3–5).

Wolff-Parkinson-White Syndrome

The Wolff-Parkinson-White (WPW) syndrome (short P–R interval and delta waves before the QRS) is characterized by recurrent supraventricular tachycardias that originate from an accessory pathway in the AV node. (The mechanism for these tachycardias is explained in the section on re-entrant tachycardias.) When atrial fibrillation occurs in a patient with an accessory pathway, drugs that block conduction in the AV node, (e.g., calcium channel blockers, β-blockers, digoxin) are unlikely to slow the ventricular rate because the accessory pathway is not blocked. Furthermore, selective block of the AV node can precipitate ventricular fibrillation (6). Therefore, *drugs that block the AV node should NOT be used when AF is associated with the WPW syndrome*. The preferred management in this situation is electrical cardioversion, or pharmacological cardioversion with amiodarone or procainamide.

MULTIFOCAL ATRIAL TACHYCARDIA

Multifocal atrial tachycardia or MAT (see Panel A in Figure 19.3) is a disorder of the elderly (average age = 70), and over half of the cases occur in patients with chronic lung disease (30). Other associated conditions include magnesium and potassium depletion, and coronary artery disease (31).

Acute Management

The following measures are recommended for the acute management of MAT, but this is a stubborn arrhythmia that is often refractory to medical management.

. Correct hypomagnesemia and hypokalemia if necessary. If both electrolyte abnormalities co-exist, the magnesium deficiency must be corrected before potassium deficits can be replaced. This is explained in Chapter 37.

. Since serum magnesium levels can be normal when total body magnesium is depleted (also explained in Chapter 37), intravenous magnesium can be given empirically using the following regimen: *Start with 2 grams $MgSO_4$ (in 50 mL saline) IV over 15 minutes, then infuse 6 grams $MgSO_4$ (in 500 mL saline) over 6 hours*. This regimen had a remarkable 88% success rate in converting MAT to a sinus rhythm in one study (31), and the effect was independent of serum magnesium level.

. If the prior measures fail, and COPD is not the cause of the MAT, *metoprolol in the doses shown in Table 19.1 has a reported 80% success rate in converting MAT to sinus rhythm* (30).

. If metoprolol is a concern in patients with COPD, the calcium channel blocker *verapamil* can be effective. Verapamil converts MAT to sinus rhythm in less than 50% of cases (30), but it can slow the ventricular rate. *The dose is 0.25–5 mg IV over 2 min, which can be repeated every 15–30 min, if necessary, to a total dose of 20 mg* (32). Verapamil is a potent negative inotropic agent, and is not recommended for patients with heart failure and a reduced ejection fraction (32).

PAROXYSMAL SUPRAVENTRICULAR TACHYCARDIAS

Paroxysmal supraventricular tachycardias (PSVT) are narrow-QRS-complex tachycardias that are second only to atrial fibrillation as the most prevalent rhythm disturbances in the general population.

Mechanism

These arrhythmias occur when impulse transmission in one pathway of the AV conduction system is slowed. This creates a difference in the refractory period for impulse transmission in the abnormal and normal conduction pathways, and this allows impulses travelling down one pathway to travel back through the other pathway. The retrograde transmission of impulses is called *re-entry*, and it results in a circular pattern of impulse transmission that is self-sustaining; i.e., a *re-entrant tachycardia*. Re-entry is triggered by an ectopic atrial impulse in one of the two conduction pathways, which results in the abrupt onset that is characteristic of re-entrant tachycardias.

There are 5 different types of PSVT, based on the location of the re-entrant pathway. The most common PSVT is *AV nodal re-entrant tachycardia*, where the re-entrant pathway is located in the AV node.

AV Nodal Re-Entrant Tachycardia

AV nodal re-entrant tachycardia (AVNRT) accounts for 50 to 60% of cases of PSVT (33). It typically appears in patients who have no history of heart disease, and is more common in women than men. The onset is abrupt, and the predominant complaints are palpitations and lightheadedness. There is no evidence of heart failure or myocardial ischemia, and significant hemodynamic compromise is uncommon. The ECG shows a narrow- QRS-complex tachycardia with a regular rhythm and a heart rate between 140 and 250 bpm (33). There are often no visible P waves on the ECG, as shown in Figure 19.2.

Because AVNRT has a regular rhythm, it is often mistaken for sinus tachycardia. However, the onset is abrupt in AVNRT and gradual in sinus tachycardia, while the ECG shows no visible P waves in AVNRT, but there are P waves before each QRS complex in sinus tachycardia. A 12-lead ECG is often necessary for identifying P waves (which are usually most prominent in the anterior precordial leads and inferior limb leads).

Vagal Maneuvers

Maneuvers that increase vagal tone are recommended as an initial attempt to terminate AVNRT (34). A variety of maneuvers have been identified (34), including (with the patient in the supine position) carotid sinus massage (only when there are no carotid bruits), the Valsalva maneuver (maximal expiratory effort with a closed glottis), the Mueller maneuver (maximal inspiratory effort with a closed glottis), induced gag reflex (by stimulating the posterior oropharynx with a tongue depressor), and application of an ice-cold, wet towel to the face (in an attempt to elicit a diving reflex).

Vagal maneuvers have had limited success in terminating AVNRTs. In one prospective study, the conventional Valsalva maneuver and carotid sinus massage had success rates of 24% and 9%, respectively (35). However in the same study, a modified Valsalva maneuver (performed by elevating the patients legs after the standard Valsalva maneuver) had a success rate of 44% (35).

Adenosine

When vagal maneuvers are ineffective, *adenosine is the drug of choice for terminating AVNRT* (34). Adenosine is an endogenous nucleotide that relaxes vascular smooth muscle and slows conduction in the AV node. When given by rapid intravenous injection, adenosine has a rapid onset of action (<30 sec) and produces a transient AV block that can terminate AV nodal re-entrant tachycardias. Adenosine is quickly cleared from the bloodstream (by receptors on RBCs and endothelial cells), and the effects last only 1–2 minutes.

TABLE 19.4 Intravenous Adenosine for Paroxysmal SVT

Feature	Recommendations
Dosing Regimen	1. Deliver through a peripheral vein. 2. Give 6 mg by rapid IV injection and flush catheter with saline. 3. If response inadequate after 2 min, give 12 mg by rapid IV injection and flush catheter with saline. 4. If response still inadequate after 2 min, another 12 mg can be given by rapid IV injection.
Dose Adjustments	Decrease dose by 50% for: • Drug delivery into the superior vena cava • Patient receiving calcium channel blocker, β-blocker, or dipyridamole.
Drug Interactions	• Dipyridamole (blocks adenosine uptake) • Theophylline (blocks adenosine receptors)
Adverse Effects	• Bradycardia, AV Block (50%) • Facial flushing (20%) • Dyspnea (12%) • Chest Pressure (7%)
Contraindications	• Asthma • 2nd or 3rd⁰ AV block • Sick sinus syndrome

From References 34,36, and 37.

Dosing Considerations

The dosing regimen for adenosine is shown in Table 19.4 (34,36,37). The initial dose is 6 mg, which is injected rapidly into a peripheral vein and followed by a saline flush. Optimal results are obtained if the drug is injected at the hub of the catheter. If conversion to sinus rhythm does not occur after 2 minutes, a second 12 mg dose is given, and this can be repeated once if necessary. This regimen *terminates re-entrant tachycardias in over 90% of cases* (34).

The effective dose of adenosine for terminating AVNRT was determined in studies that used drug injection into peripheral veins. However, ventricular asystole has been reported when standard doses of adenosine are given through central venous catheters (38), so *a dose reduction of 50% has been recommended* by some (including the manufacturer) *when adenosine is delivered through a central venous catheter* (38).

Adverse Effects

The adverse effects of adenosine are short-lived. The most frequent adverse effect is post-conversion bradycardia, including various degrees of AV block. The AV block is refractory to atropine, but resolves spontaneously within 60 seconds (37). Dipyridamole enhances the AV block produced by adenosine, and the two drugs should not be used together (37). Adenosine is contraindicated in patients with asthma; i.e., there is evidence that adenosine produces a sense of dyspnea, but not bronchospasm, in asthmatic subjects (39).

Other Therapies

When PSVT does not respond to adenosine, the calcium channel blockers diltiazem or verapamil can be effective. (The dosing regimen for diltiazem is shown in Table 19.1, and the dosing regimen for verapamil is presented in the section on multifocal atrial tachycardia). For the rare case of PSVT that is resistant to drugs or is hemodynamically unstable, synchronized electrical cardioversion will terminate the rhythm, but energy levels greater than 100 J may be necessary (34).

VENTRICULAR TACHYCARDIA

Ventricular tachycardia (VT) is a wide-QRS-complex tachycardia that has an abrupt onset, a regular rhythm, and a rate that is typically 140–200 bpm. The appearance can be *monomorphic* (uniform QRS complexes) or *polymorphic* (multiple QRS morphologies). VT rarely occurs in the absence of structural heart disease (2), and when it is sustained (i.e., lasts longer than 30 seconds), it can be an immediate threat to life.

VT versus SVT

Monomorphic VT can be difficult to distinguish from an SVT with prolonged AV conduction. This is demonstrated in Figure 19.6. The tracing in the upper panel shows a wide-QRS-complex tachycardia that looks like monomorphic VT. The tracing in the lower panel shows spontaneous conversion to sinus rhythm. Note that the QRS complex remains unchanged after the arrhythmia is terminated, revealing an underlying bundle branch block. Thus, the apparent VT in the upper panel is actually an SVT with a pre-existing bundle branch block.

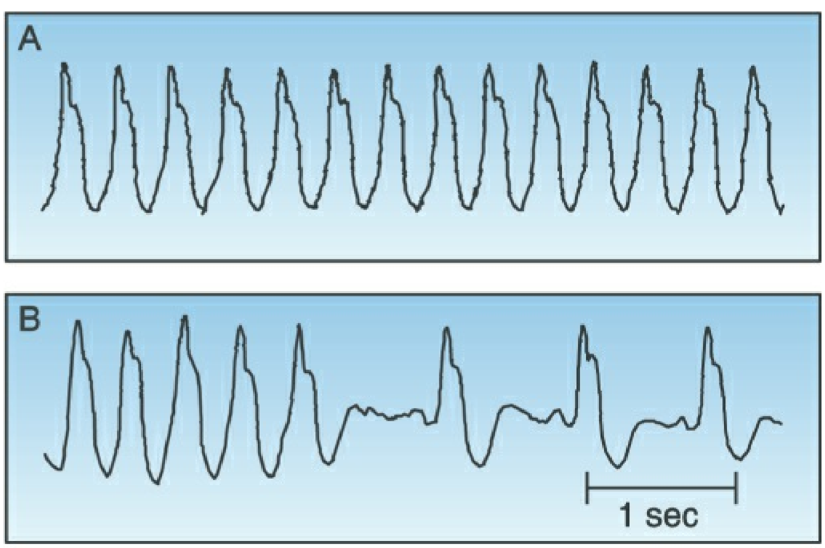

FIGURE 19.6 Upper panel shows a wide-QRS-complex tachycardia that looks like monomorphic VT. However, the lower panel shows spontaneous conversion to a sinus rhythm,

which reveals an underlying bundle branch block, indicating that the rhythm in the upper panel is an SVT with a pre-existing bundle branch block. Tracings courtesy of Dr. Richard M. Greenberg, MD.

Clues

There are two abnormalities on an ECG that will identify VT as the cause of a wide-QRS-complex tachycardia.

. The atria and ventricles beat independently in VT, and this results in *AV dissociation* on the ECG, where there is no fixed relationship between P waves and QRS complexes. (P waves are most visible in the inferior limb leads and the anterior precordial leads.)

. The presence of fusion beats like the one in Figure 19.7 is indirect evidence of VT. A fusion beat is produced by the retrograde transmission of a ventricular ectopic impulse that collides with a supraventricular (e.g., sinus node) impulse. The result is a hybrid QRS complex that is a mixture of the normal QRS complex and the ventricular ectopic impulse. The presence of a fusion beat (which should be evident on a single-lead ECG tracing) is thus indirect evidence of ventricular ectopic activity.

If there is no definitive evidence of VT on the ECG, the presence or absence of heart disease can be useful; i.e., *VT is the cause of 95% of wide-QRS-complex tachycardias in patients with underlying heart disease* (40). Therefore, a wide-QRS-complex tachycardia should be treated as probable VT in any patient with underlying heart disease.

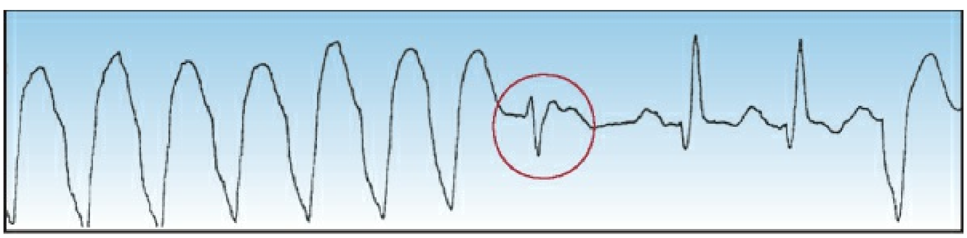

FIGURE 19.7 An example of a fusion beat (circled in red), which is a hybrid QRS complex produced by the collision of a ventricular ectopic impulse and a supraventricular (e.g., sinus node) impulse. The presence of fusion beats is evidence of ventricular ectopic activity.

Management

The management of patients with a wide-QRS-complex tachycardia can be organized as shown in Figure 19.8.

. If there is evidence of hemodynamic compromise, *electrical cardioversion* is the appropriate intervention, regardless of whether the rhythm is VT or SVT with aberrant conduction. The shocks should be synchronized (timed with the QRS complex) and a low-energy shock of 100 J (biphasic or monophasic shocks) should terminate most cases of monomorphic VT (41). If this is unsuccessful, try 100 J again but use biphasic shocks. You can escalate to 200 J, but avoid higher energy shocks (e.g., 360 J) because they damage the myocardium (41).

. If there is no evidence of hemodynamic compromise and the diagnosis of VT is certain, intravenous amiodarone should be used to terminate the arrhythmia. *Amiodarone is the favored drug for suppressing monomorphic VT* (4), and the dosing regimen is shown in Figure 19.8.

. If there is no evidence of hemodynamic compromise and the diagnosis of VT is uncertain, the response to adenosine can be helpful because adenosine will abruptly terminate most cases of paroxysmal SVT, but will not terminate VT. If a wide-QRS-complex tachycardia is refractory to adenosine, the likely diagnosis is VT, and intravenous amiodarone is indicated for arrhythmia suppression.

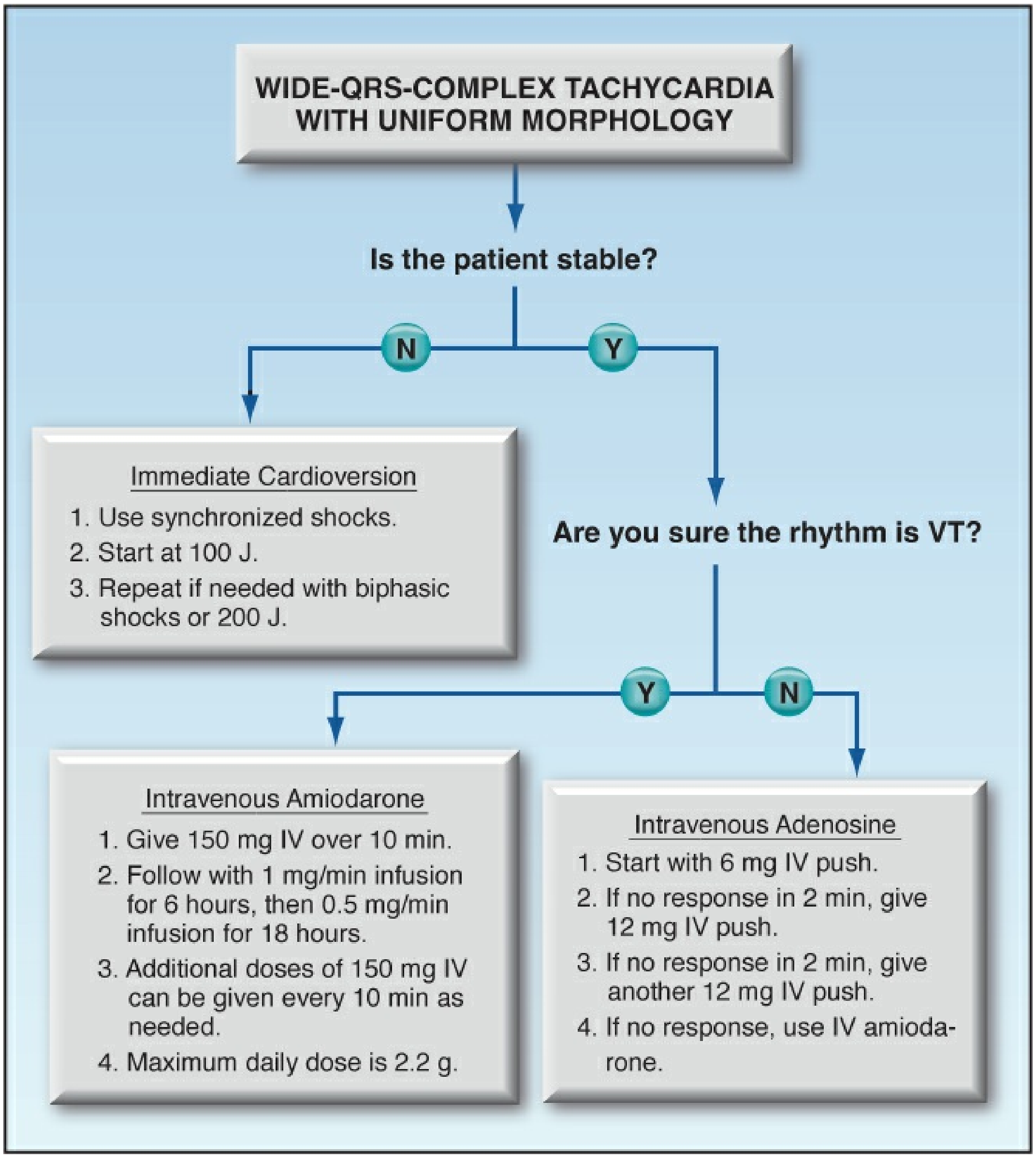

FIGURE 19.8 Flow diagram for the acute management of patients with wide-QRS-complex tachycardia. Based on recommendations in Reference 4.

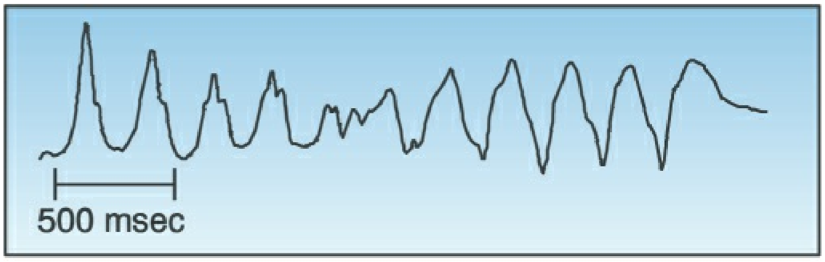

FIGURE 19.9 Torsade de pointes, a polymorphic ventricular tachycardia described as “twisting around the (isoelectric) points”. Tracing courtesy of Dr. Richard M. Greenberg, MD.

Torsade de Pointes

Torsade de pointes ("twisting around the points") is a polymorphic VT with QRS complexes that appear to be twisting around the isoelectric line of the ECG, as shown in Figure 19.9. This arrhythmia is usually associated with a prolonged QT interval, and it can be congenital or acquired. The acquired form is much more prevalent, and is caused by a variety of drugs and electrolyte abnormalities that prolong the QT interval (42,43). There is also a polymorphic VT that is associated with a normal QT interval, and the predisposing condition in this case is myocardial ischemia (42).

Predisposing Factors

The drugs most frequently implicated in torsade de pointes are listed in Table 19.5 (43). The prominent offenders are antiarrhythmic agents (Class IA and III), macrolide antibiotics, neuroleptic agents (phenothiazines and butyrophenones), and cisapride (a pro-motility drug). The electrolyte disorders that prolong the QT interval include hypokalemia, hypocalcemia, and hypomagnesemia.

TABLE 19.5 Drugs That Can Induce Torsade de Pointes

Antiarrhythmics		Antimicrobials	Neuroleptics	Others
IA	Quinidine	Clarithromycin	Chlorpromazine	Cisapride
	Disopyramide	Erythromycin	Tioridazine	Methadone
	Procainamide	Pentamidine	Droperidol	
			Haloperidol	
III	Ibutilide			
	Sotalol			

From Reference 43. For a complete list of drugs, go to www.torsades.org

Measuring the QT Interval

The QT interval is the ECG manifestation of ventricular depolarization and repolarization, and is measured from the onset of the QRS complex to the end of the T wave. The longest QT intervals are usually in precordial leads V3 and V4, and these leads are most reliable for assessing QT prolongation. The QT interval varies inversely with heart rate, and a rate-corrected QT interval (QTc) provides a more accurate assessment of QT prolongation. The accepted method for determining the QTc is to divide the QT interval by the square root of the R-R interval (43–45); i.e.,

$$QTc = QT/\sqrt{R\text{-}R} \tag{19.1}$$

A normal QTc is ≤0.44 seconds, and a QTc >0.5 seconds represents a risk for torsade de pointes (45). However, the importance of QT prolongation as a risk factor for torsade de pointes is not as advertised, because prolonged QT intervals are commonplace in ICU patients, while torsade de pointes is an uncommon arrhythmia.

Management

The management of polymorphic VT can be summarized as follows:

- Sustained polymorphic VT requires nonsynchronized electrical cardioversion (i.e., defibrillation) (42).
- Intravenous magnesium is popular for terminating torsade de pointes, and can be combined with electrical cardioversion (46). There is no consensus about the magnesium dose in this setting, but the current ACLS guidelines recommend 1–2 grams of magnesium sulfate ($MgSO_4$) IV over 15 minutes (6).
- Other measures aimed at preventing recurrences of torsade de pointes include correcting hypokalemia and hypomagnesemia (high-risk electrolyte abnormalities) and discontinuing high-risk drugs (e.g., haloperidol).
- For polymorphic VT with a normal QT interval, amiodarone or β-blockers can help to prevent recurrences (47).

A FINAL WORD

Managing the Muddle

The first glance at the variety of cardiac arrhythmias and treatment options is often a befuddling experience, and the following tips can help to unravel things.

For a narrow-complex tachycardia:

- If the rhythm is regular, the candidates are sinus tachycardia, paroxysmal supraventricular tachycardia (PSVT), or atrial flutter. However, PSVT is uncommon in ICU patients (and usually occurs in younger patients with no structural heart disease).
- If the rhythm is highly irregular, it is most likely atrial fibrillation. Another candidate is multifocal atrial tachycardia (MAT), which is seen in patients with chronic lung disease.
- Be aware that rapid atrial fibrillation can appear to be a regular rhythm.

For a wide-complex tachycardia:

- If the rhythm is irregular, it is likely to be atrial fibrillation (or MAT) with prolonged AV conduction. Ventricular tachycardia is never irregular.

. If the rhythm is regular, it is most likely ventricular tachycardia. However, rapid atrial fibrillation with prolonged AV conduction can also appear to be regular.

. If the height of the complexes is increasing and decreasing, the rhythm is likely to be "torsade de pointes".

Finally, atrial fibrillation is (by far) the most common arrhythmia you will encounter in the ICU, so focus your attention accordingly.

References

a. Osler W. On the educational value of the medical society. Yale Medical Journal, 1903.

Guidelines and Reviews

1. Al-Salamah T, AlAgeel M, Alblaihed L. Narrow Complex Tachycardias. Emerg Med Clin N Am 2022; 40:717–732.
2. Alblaihed L, Al-Salamah T. Wide-complex tachycardias. Emerg Med Clin N Am 2022; 40:733–753.
3. Hindricks G, Potpara T, Dagres N, et al. 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation developed in collaboration with the European Association of Cardio-Thoracic Surgery (EACTS). Eur Heart J 2020; 42:373–498.
4. January CT, Wann LS, Calkins H, et al. 2019 AHA/ACC/HRS focused update of the 2014 AHA/ACC/HRS guideline for the management of patients with atrial fibrillation: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation. 2019; 140:e125–e151.
5. January CT, Wann SL, Alpert JS, et al. 2014 AHA/ACC/HRS guidelines for the management of patients with atrial fibrillation: executive summary. J Am Coll Cardiol 2014; 64:2246–2280.
6. Panchal AR, Bartos JA, Carbañas JG, et al. Part 3: Adult basic and advanced life support: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2020; 142(Suppl 2):S366–S468.

Atrial Fibrillation

7. Wetterslev M, Haase N, Hassager C, et al. New-onset atrial fibrillation in adult critically ill patients: a scoping review. Intensive Care Med 2019; 45:928–938.
8. Bedford JP, Ferrando-Vivas P, Redfern O, et al. New-onset atrial fibrillation in intensive care: epidemiology and outcomes. Eur Heart J Acute Cardiovasc Care 2022; 11:620–628.
9. Mayson SE, Greenspon AJ, Adams S, et al. The changing face of postoperative atrial fibrillation: a review of current medical therapy. Cardiol Rev 2007; 15:231–241.
10. Davis EM, Packard KA, Hilleman DE. Pharmacologic prophylaxis of postoperative atrial fibrillation in patients undergoing cardiac surgery: beyond beta blockers. Pharmacotherapy 2010; 30:274e–318e.
11. Guyton AC. The relationship of cardiac output and arterial pressure control. Circulation 1981; 64:1079–1088.
12. Karadavut S, Altintop I. Pulse deficit in atrial fibrillation—a different perspective on rhythm or rate control strategy. Kardiol Pol 2021; 79:1231–1238.
13. Drugs for atrial fibrillation. Med Lett Drugs Ther 2019; 61:137–144.
14. Posen A, Bursua A, Petzel R. Dosing strategy effectiveness of diltiazem in atrial fibrillation with rapid ventricular response. Ann Emerg Med 2023; 81:288–296.
15. Ellenbogen KA, Dias VC, Cardello FP, et al. Safety and efficacy of intravenous diltiazem in atrial fibrillation or atrial flutter. Am J Cardiol. 1995; 75:45–49.
16. Siu C-W, Lau C-P, Lee W-L, et al. Intravenous diltiazem is superior to intravenous amiodarone or digoxin for achieving ventricular rate control in patients with acute uncomplicated atrial fibrillation. Crit Care med 2009; 37:2174–2179.
17. Karth GD, Geppert A, Neunteufl T, et al. Amiodarone versus diltiazem for rate control in critically ill patients with atrial tachyarrhythmias. Crit Care Med 2001; 29:1149–1153.
18. Olshansky B, RosenfeldLE, Warner AI, et al. The Atrial Fibrillation Follow-up Investigation of Rhythm Management (AFFIRM) study: approaches to control rate in atrial fibrillation. J Am Coll Cardiol 2004; 43:1201–1208.
19. Gray RJ. Managing critically ill patients with esmolol. An ultra short acting β-adrenergic blocker. Chest 1988; 93:398–404.
20. Clemo HF, Wood MA, Gilligan DM, Ellenbogen KA. Intravenous amiodarone for acute heart rate control in the critically ill patient with atrial tachyarrhythmias. Am J Cardiol. 1998; 81:594–598.
21. Gallik DM, Singer I, Meissner MD, et al. Hemodynamic and surface electrocardiographic effects of a new aqueous formulation of intravenous amiodarone. Am J Cardiol 2002; 90:964–968.
22. Giardina E-G, Passman R. Amiodarone: clinical uses. UpToDate (www.uptodate.com). Accessed 10/20/2023.

23. Niebauer MJ, Brewer JE, Chung MK, Tchou PJ. Comparison of the rectilinear biphasic waveform with the monophasic damped sine waveform for external cardioversion of atrial fibrillation and flutter. Am J Cardiol 2004; 93:1495–1499.
24. Man-Son-Hing M, Nichol G, Lau A, Laupacis A. Choosing antithrombotic therapy for elderly patients with atrial fibrillation who are at risk for falls. Arch Intern Med 1999; 159:677–685.
25. Hart RG, Pearce LA, Aguilar MI. Meta-analysis: antithrombotic therapy to prevent stroke in patients who have nonvalvular atrial fibrillation. Ann Intern Med 2007; 146:857–867.
26. Connolly SJ, Ezekowitz MD, Yusuf S, et al; for the RE-LY Steering Committee Investigators. Dabigatran versus warfarin in patients with atrial fibrillation. N Engl J Med 2009; 361:1139–1151.
27. Patel MR, Mahaffey KW, Garg J, et al; for the ROCKET AF Investigators. Rivaroxaban versus warfarin in nonvalvular atrial fibrillation. N Engl J Med 2011; 365:883–891.
28. Granger CB, Alexander JH, McMurray JJ, et al; for the ARISTOTLE Committees and Investigators. Apixaban versus warfarin in patients with atrial fibrillation. N Engl J Med 2011; 365:981–992.
29. Giugliano RP, Ruff CT, Braunwald E, et al; for the ENGAGE AF-TIMI Investigators. Edoxaban versus warfarin in patients with atrial fibrillation. N Engl J Med 2013; 369:2093–2104.

Multifocal Atrial Tachycardia

30. Kastor J. Multifocal atrial tachycardia. N Engl J Med 1990; 322:1713–1720.
31. Iseri LT, Fairshter RD, Hardeman JL, Brodsky MA. Magnesium and potassium therapy in multifocal atrial tachycardia. Am Heart J 1985; 312:21–26.
32. Neumar RW, Otto CW, Link MS, et al. Part 8: adult advanced cardiovascular life support. 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2010; 122(Suppl):S729–S767.

Paroxysmal Supraventricular Tachycardias

33. Trohman RG. Supraventricular tachycardia: implications for the internist. Crit Care Med 2000; 28(Suppl):N129–N135.
34. Page RL, Joglar JA, Caldwell MA, et al. 2015 ACC/AHA/HRS guideline for the management of adult patients with supraventricular tachycardia: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation 2015; 133:e507–e574.
35. Ceylan E, Ozpolat C, Onur O, et al. Initial and sustained response effects of 3 vagal maneuvers in supraventricular tachycardia: A randomized, clinical trial. J Emerg Med 2019; 57:299–305.
36. Rankin AC, Brooks R, Ruskin JM, McGovern BA. Adenosine and the treatment of supra-ventricular tachycardia. Am J Med 1992; 92:655–664.
37. Chronister C. Clinical management of supraventricular tachycardia with adenosine. Am J Crit Care 1993; 2:41–47.
38. McCollam PL, Uber W, Van Bakel AB. Adenosine related ventricular asystole. Ann Intern Med 1993; 118:315–316.
39. Burki NK, Alam M, Lee L-Y. The pulmonary effects of intravenous adenosine in asthmatic subjects. Respir Res 2006; 7:139–146.

Ventricular Tachycardia

40. Akhtar M, Shenasa M, Jazayeri M, et al. Wide QRS complex tachycardia. Ann Intern Med 1988; 109:905–912.
41. Link MS, Atkins DL, Passman RS, et al. Part 6: electrical therapies: automated external defibrillators, defibrillation, cardioversion, and pacing: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation. 2010; 122(18 Suppl 3): S706–S719.
42. Vukmir RB. Torsade de pointes: a review. Am J Emerg Med 1991; 9:250–262.
43. Gupta A, Lawrence AT, Krishnan K, et al. Current concepts in the mechanism and management of drug-induced QT prolongation and torsade de pointes. Am Heart J 2007; 153:891–899.
44. Sadanaga T, Sadanaga F, Yoo H, et al. An evaluation of ECG leads used to assess QT prolongation. Cardiology 2006; 105:149–154.
45. Al-Khatib SM, LaPointe NM, Kramer JM, Califf RM. What clinicians should know about the QT interval. JAMA 2003; 289:2120–2127.
46. McLean RM. Magnesium and its therapeutic uses: a review. Am J Med 1994; 96:63–76.
47. Gupta AK, Thakur RK. Wide QRS complex tachycardias. Med Clin N Am 2001; 85:245–266.

Chapter 20

Acute Coronary Syndromes

The study of the causes of things must be preceded by the study of things caused.

John Hughlings Jackson (1835–1911)

One of the seminal moments in cardiac critical care was the discovery (in 1980) that transmural myocardial infarction (MI) was the result of an occlusive thrombus in a coronary artery. This led to the introduction of thrombolytic therapy (in the mid-1980s), followed by the superseding technology of balloon angioplasty and stent placement. Advances in reperfusion therapy have improved outcomes, and have contributed to 20% decline in the annual mortality rate from coronary disease over the past decade (from 2010 to 2020) (1). However, coronary artery disease continues to be the leading cause of death in the United States (1), so there is more work to be done.

This chapter describes the diagnosis and early management of acute myocardial infarction and unstable angina (the *acute coronary syndromes*), with an emphasis on management in the first 24–36 hours after presentation. Also included is a section on acute aortic dissection, which has a clinical presentation that can be confused with acute coronary syndromes. Many of the recommendations in this chapter are derived from the clinical practice guidelines listed in the bibliography at the end of the chapter (2–7).

CORONARY THROMBOSIS

Pathogenesis

As just mentioned, acute myocardial infarction is the result of an occlusive thrombus in one or more coronary arteries. The trigger for thrombus formation is rupture of an atherosclerotic plaque, which releases thrombogenic lipids (see Figure 20.1). Plaque disruption is attributed to inflammation (8), but hydraulic shear stress may also play a role, because ruptured plaques are typically located at branch points in the coronary circulation (9).

Role of Oxidative Injury

Inflammation plays a major role in both the genesis and rupture of atherosclerotic plaques (10). The damaging effects of inflammation are largely due to the oxidative actions of “reactive oxygen species”, as described in Chapter 17 (see Figure 17.1), and this implicates oxidative injury as a major participant in coronary artery disease. The principle culprit in this scenario appears to be *myeloperoxidase* (MPO), an enzyme released by leukocytes that generates hypochlorite (the active ingredient in household bleach), and oxidizes lipoproteins in atherosclerotic plaques (11). Clinical studies have shown a direct correlation between plasma levels of MPO and the presence and severity of coronary artery disease (12,13), and autopsy studies in humans have shown extensive staining for MPO at the rupture site of coronary artery plaques (13). These observations suggest that inhibition of MPO is a potential treatment modality for coronary artery disease.

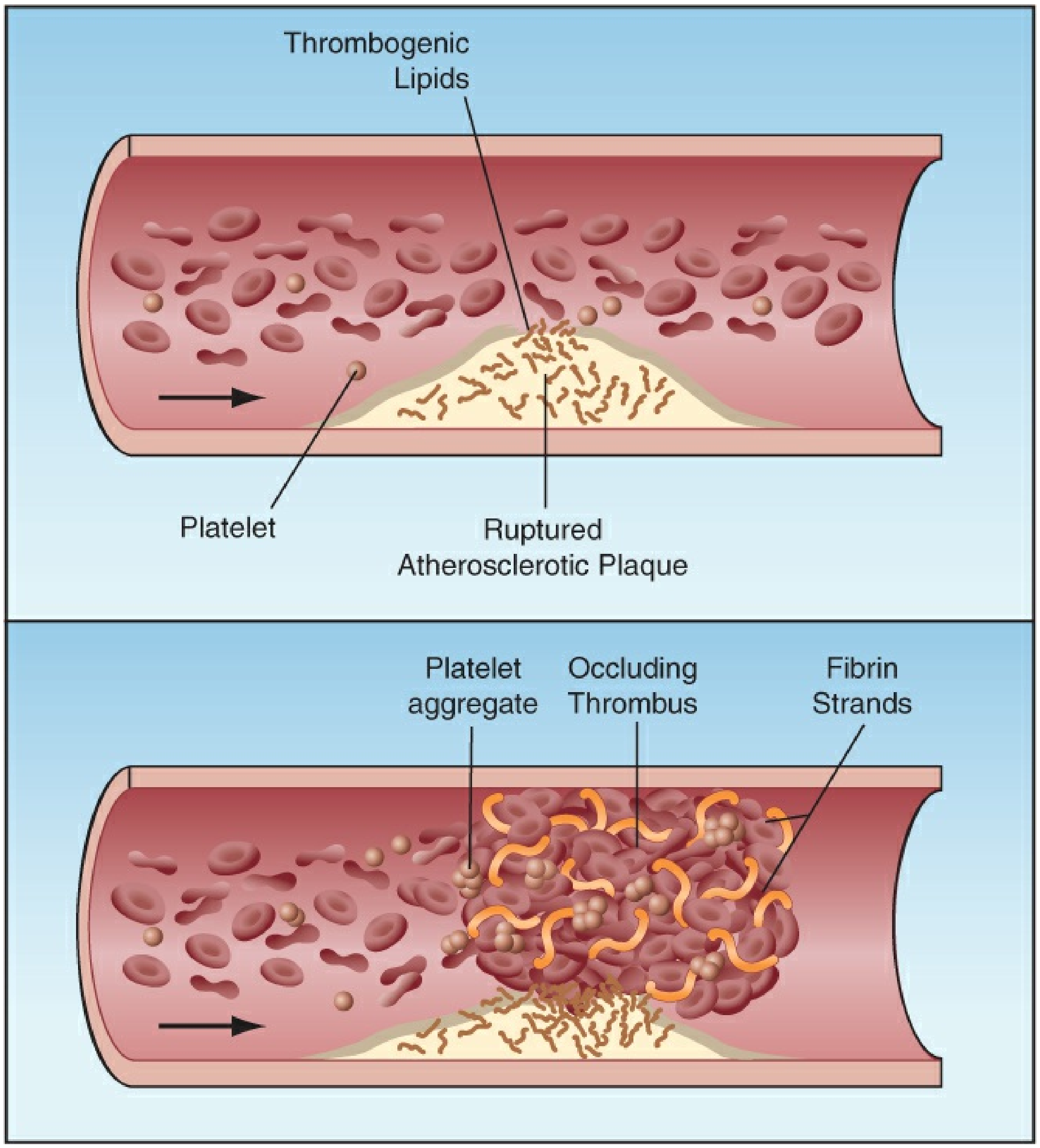

FIGURE 20.1 Pathogenesis of acute coronary syndromes. Rupture of an atherosclerotic plaque releases thrombogenic lipids that activate platelets and clotting factors (upper panel), resulting in

the formation of an occlusive thrombus (lower panel).

Clinical Syndromes

Acute coronary thrombosis produces three distinct clinical syndromes (which are the acute coronary syndromes, or ACS), which are classified by the presence or absence of ST-segment elevation on a 12-lead electrocardiogram (ECG):

. ST-elevation myocardial infarction (STEMI), which is a transmural infarction caused by complete occlusion of the infarct-related artery.

. Non-ST-elevation myocardial infarction (NSTEMI), which is the result of incomplete or partial occlusion of the infarct-related artery.

. Unstable angina (UA), which is not associated with ST elevation on the ECG, and is the result of repetitive on-off episodes of coronary occlusion.

NSTEMI and UA are typically grouped together as “non-ST-elevation acute coronary syndromes”, or NSTE-ACS.

Diagnostic Evaluation

The success of treating ACS is time-dependent, so the diagnostic evaluation must be completed as quickly as possible, The evaluation has three components: the clinical presentation, the 12-lead ECG, and the high-sensitivity troponin assay.

CLINICAL PRESENTATION: The clinical presentation of ACS can vary widely, but about 80% of patients have some type of chest discomfort (e.g., pain, pressure, tightness, etc.) (3,4). The onset is abrupt, and lasts for longer than 15 minutes. The discomfort is a deep sensation that cannot be localized to a specific point, and can involve the shoulders, neck, arms, and abdomen. Patients are often apprehensive, and can complain of nausea, vomiting, and dyspnea.

ACS is unlikely when chest pain is sharp or fleeting, or can be localized to a precise point, and also if the pain occurs during inspiration, or can be elicited by body movements or chest percussion. Of particular note, *chest pain that is relieved by nitroglycerin is not evidence of myocardial ischemia, since nitroglycerin can also relieve chest pain from esophageal spasm* (14).

The initial evaluation may also reveal evidence of acute heart failure or cardiogenic shock. These complications are not considered in this chapter, but are described in detail in Chapters 16 and 18.

ELECTROCARDIOGRAM: A 12-lead electrocardiogram (ECG) *should be obtained within 10 minutes of the first patient contact* (2). Some of the ECG changes in ACS are shown in Table 20.1 ECG evidence of ST-elevation MI (STEMI) merits emergency reperfusion therapy (described later). NSTEMI and UA (NSTE-ACS) may be accompanied by ST depressions or T-wave inversions, but the ECG can also be normal in these conditions. Finally, ischemic changes in the ST segment can be masked by a bundle branch block or a paced rhythm, and when coronary ischemia is suspected, the presence of these ECG abnormalities is managed like a STEMI (i.e., with emergent reperfusion therapy).

TROPONIN ASSAY: Plasma levels of cardiac troponin (cTn) are used to detect the presence of myocardial cell injury, and a high-sensitivity assay (hs-cTn) can detect elevations of plasma cTn as early as 3 hours after the onset of symptoms (15). There are several commercial hs-cTn assays, and the normal or reference levels differ for each assay, and can differ for each hospital. The following protocol is recommended for hs-cTn levels (2–6).

- Plasma levels of hs-cTn are measured at the time of presentation, and again one hour later.
- Myocardial necrosis is likely if the initial hs-cTn level is elevated (usually above the 99% upper reference level for the assay), and there has been at least 3 hours since the onset of symptoms. Conditions other than ischemia can cause myocardial injury, hence a repeat hs-cTn level at one hour is used to determine if the myocardial injury is ischemic in origin (see next).
- A significant change (usually >10%) in the second hs-cTn level is evidence of acute ischemia (i.e., acute MI).

There are several non-ischemic causes of elevated hs-cTn levels (e.g., cardiomyopathy, sustained tachycardia, heart failure, pulmonary hypertension, and even sepsis) (2,3) but these conditions should not cause an acute change in the hs-cTn levels. However, patients with markedly elevated hs-cTn levels are usually admitted to the hospital for further testing.

TABLE 20.1 ECG Changes in Acute Coronary Syndromes

Condition	Pattern	Criteria
STEMI		ST elevation at the J point in ≥2 contiguous leads: either ≥1.5 mm (females) or ≥2.0 mm (males) in V_2–V_3, or ≥1 mm in the other leads.
Posterior STEMI	 $V_1 – V_3$	ST depression in V_1–V_3 with a positive T wave.
NSTE-ACS		J point depression ≥0.5 mm in V_2–V_3, or ≥1 mm in all other leads. ST segment can be horizontal or downsloping.
NSTE-ACS		T wave inversion >1 mm in ≥5 leads, including I, II, aVL, and V_2–V_6.

From Reference 5.

REPERFUSION STRATEGIES

When the likelihood of an acute coronary occlusion is very high using the evaluation just described, the next step is to determine the appropriate strategy for restoring flow in the occluded artery. Three methods are available for coronary reperfusion:

- Percutaneous coronary intervention (PCI), which involves coronary angiography (to identify the obstruction), balloon angioplasty (to restore patency), and placement of a stent (to prevent re-occlusion). This is the standard reperfusion method.
- Thrombolytic drug therapy, which is a less effective alternative when PCI is not available.
- Coronary artery bypass surgery, which is reserved for cases where PCI is not warranted, or is not successful.

Strategies

The reperfusion strategies for ACS are based on the presence or absence of ST-elevation on the ECG, and the availability of PCI. The following recommendation are from the most recent clinical practice guidelines (2,3,5–7).

ST-Elevation MI

The management outlined below is for patients with ECG evidence of STEMI, and for cases where myocardial ischemia is likely and there is a bundle branch block on the ECG.

- The optimal management is emergency PCI, performed within 90–120 minutes of hospital arrival (i.e., the “door-to-balloon” time) (2,6).
- If PCI is unavailable, patients should be transferred immediately to a PCI-capable hospital, with a target door-to-balloon time of 120 minutes. If the transfer time is expected to exceed 120 minutes, thrombolytic therapy can be given prior to the transfer, and should be initiated within 30 minutes of hospital arrival (i.e., the “door-to-needle” time) (6).
- Coronary artery bypass surgery is reserved for cases where PCI fails to establish reperfusion and there is ongoing ischemia.

DOOR-TO-BALLOON TIME: The negative impact of delays in performing balloon angioplasty is shown in Figure 20.2 (16). Note that the mortality rate is significantly increased when the time from hospital arrival to coronary angioplasty (the “door-to-balloon” time) exceeded 120 minutes. The American Heart Association recommends 90 minutes as the target door-to-balloon time (6), which allows a 30-minute buffer period.

INTERHOSPITAL TRANSFERS: Less than 30% of hospitals in the United States are equipped to perform PCI (17), and about one of every four patients with a STEMI is brought to a non-PCI hospital (18). When this occurs, transfer to a PCI-capable hospital can provide a survival benefit if the total door-to-balloon time does not exceed 120 minutes (19). When long delays are anticipated, thrombolytic therapy can be used as a bridge to PCI (2,6).

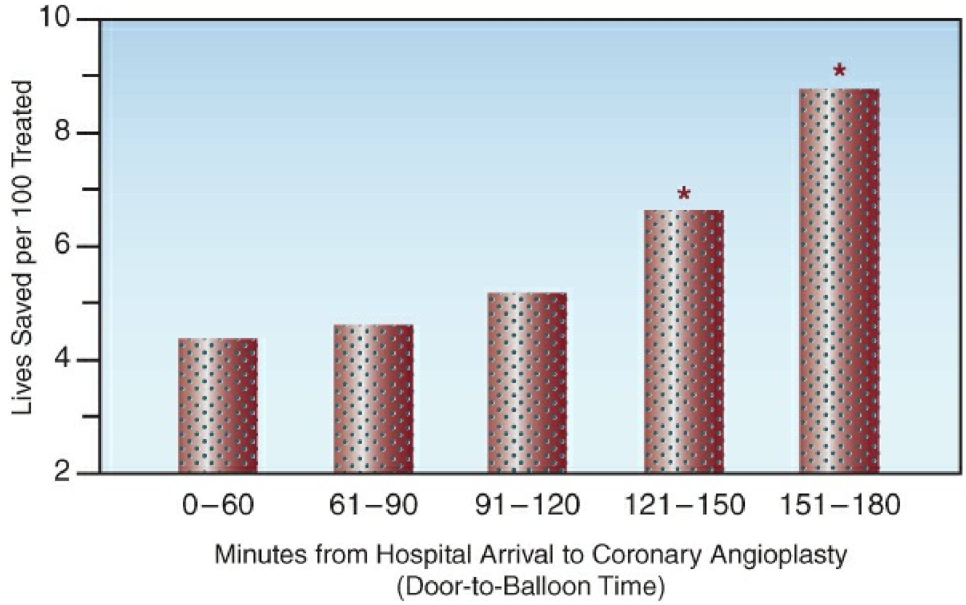

FIGURE 20.2 Mortality rate in relation to the time from hospital arrival to coronary angioplasty in patients with STEMI. Asterisks indicate a significant difference compared to the initial time period (0–60 min). Adapted from data in Reference 16.

Non-ST-Elevation ACS

- Urgent (as soon as possible) PCI is advised for cases of non-ST-elevation MI (NSTEMI) that are complicated by hemodynamic instability, cardiogenic shock, acute heart failure, or persistent chest pain (2,5). Otherwise, PCI can be performed 24–72 hours after presentation (5).
- Thrombolytic therapy has no survival benefit in NSTEMI, and is not used as an alternative to PCI.
- Patients with unstable angina who are hemodynamically stable and devoid of pain may not require PCI during hospitalization.

Thrombolytic Therapy

Thrombolytic therapy is less effective than PCI (19), and is limited to uncomplicated cases of STEMI where PCI is not immediately available. The following are additional limiting factors for thrombolytic therapy.

- The survival benefit of thrombolytic therapy is time-dependent; i.e., it is greatest in the first few hours after symptom onset, and then gradually declines and is lost after 12 hours (20). This is demonstrated in Figure 20.3, and is the basis for the recommendation that *thrombolytic therapy is not advised if more than 12 hours have elapsed from symptom onset* (6).
- In addition to time constraints, the use of thrombolytic therapy is limited by the risk of bleeding. Absolute and relative contraindications to thrombolytic therapy are presented in Chapter 47.

Thrombolytic Agents

Thrombolytic agents act by converting plasminogen to plasmin, which then breaks fibrin strands into smaller subunits. The thrombolytic agents approved for clinical use, are shown in Table 20.2. These agents act on the plasminogen that is bound to fibrin strands in the blood clot, and this limits the extent of systemic fibrinolysis, and reduces the risk of troublesome bleeding. The lytic agents in Table 20.2 are equally capable of restoring patency in occluding coronary arteries (6).

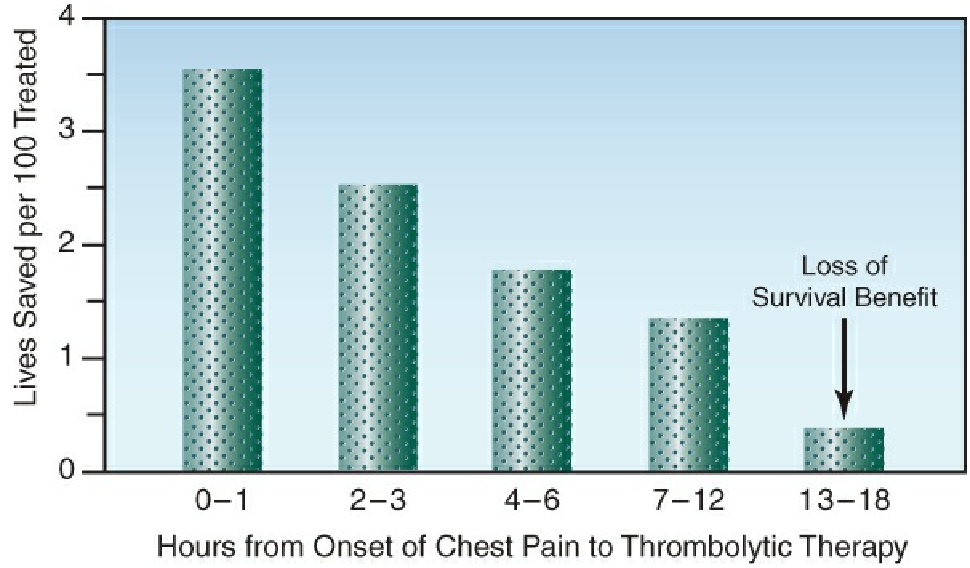

FIGURE 20.3 The survival benefit of thrombolytic therapy in relation to the time elapsed from the onset of chest pain. Data from 45,000 patients with STEMI or left bundle branch block in Reference 20.

- *Alteplase* (Activase) is a recombinant tissue plasminogen activator (tPA) that was the first clot-specific fibrinolytic agent approved for clinical use. However, it is no longer favored because it is relatively slow-acting (21), and requires 90 minutes to complete the dosing regimen.
- *Reteplase* (Retavase) is a recombinant variant of tPA that is given as two bolus doses over a 30-minute period. It produces more rapid clot lysis than alteplase (21), but clinical trials show no survival advantage (22).
- *Tenecteplase* (TNK-tPA) is another variant of tPA that is given as a single IV bolus, and produces more rapid clot lysis than reteplase (23). It is currently the most popular thrombolytic agent, but has no proven survival advantage (24).

TABLE 20.2 Thrombolytic Therapy for Acute Coronary Occlusion

Agent	Dosing Regimen	Patency Rate at 90 min.
Alteplase (tPA)	15 mg IV bolus, then 0.75 mg/kg (not >50 mg) over 30 min, then 0.5 mg/kg (not >35 mg) over 60 min. Max dose: 100 mg over 90 min.	73–84%
Reteplase (rPA)	10 Units as IV bolus and repeat in 30 min.	84%

Tenecteplase (TNK-tPA)	Single IV bolus: 30 mg for wt. <60 kg, 35 mg for 60–69 kg, 40 mg for 70–79 kg, 45 mg for 80–89 kg, and 50 mg for ≥90 kg.	85%

From Reference 6.

Major Bleeding

Thrombolytic therapy for ACS is complicated by significant bleeding (i.e., requires blood transfusions) in about 10% of cases, and life-threatening bleeding (shock or intracranial hemorrhage) occurs in about 1% of cases (25). The bleeding is associated with low plasma fibrinogen levels (usually <100 mg/dL) and can be treated with concentrated fibrinogen products such as cryoprecipitate or fibrinogen concentrates (see the last section of Chapter 13). The use of antifibrinolytic agents such as tranexamic acid is discouraged because of the risk of thrombosis (26).

CARDIOPROTECTIVE MEASURES

The following measures are aimed at improving the balance between myocardial O_2 supply and O_2 demand, and are initiated when myocardial ischemia is first suspected (sometimes in the pre-hospital setting). These measures are summarized in Table 20.3.

Oxygen

Despite a long history of routine use in ACS, supplemental O_2 provides no benefit in patients with ACS who have a normal arterial O_2 saturation (i.e., SaO_2 ≥90%) (27). As a result, the current recommendation is to *use supplemental oxygen only in patients who are hypoxemic; i.e., have an SaO_2 <90%* (2,5,6). More compelling reasons to avoid the unnecessary use of oxygen in ACS are as follows:

- Oxygen promotes vasoconstriction in the coronary arteries, and can decrease coronary blood flow in patients with coronary artery disease (28).
- Reactive oxygen species are implicated in the myocardial injury that follows coronary reperfusion (29).

(For more on the dark side of oxygen, see Chapter 25.)

TABLE 20.3 Cardioprotective Measures

Agent	Dosing Regimens and Comments
Oxygen	Dosing: Whatever is needed to maintain an SaO_2 ≥90%. Comment: Use supplemental O_2 judiciously, because it promotes coronary artery vasoconstriction.
Nitroglycerin	Dosing: For chest pain: 0.4 mg sublingual or by mouth spray every 5 min x 3, as needed. For recurrent pain, high BP or CHF: infuse at 5 µg/min initially, then titrate upward to desired end-point. Comment: Avoid in RV infarction, aortic stenosis, and for 24–48 hrs after a dose of phosphodiesterase inhibitors. Tachyphylaxis is common with prolonged (>24 hrs)

	infusions.
Morphine	Dosing: 4–8 mg IV, and follow with 2–8 mg IV every 5–15 min as needed. Comment: Reduces antiplatelet effect of P2Y12 inhibitors, but clinical significance is uncertain.
Metoprolol	Dosing: 4–8 mg IV, and follow with 2–8 mg IV every 5–15 min as needed. Comment: Avoid early use of β-blockers in acute heart failure, but not after the condition stabilizes. β-blockers contraindicated in cocaine-related ischemia.

Relieving Chest Pain

Relieving chest pain helps to alleviate unwanted cardiac stimulation from anxiety-induced adrenergic hyperactivity.

Nitroglycerin

Nitroglycerin is given as a sublingual tablet or aerosol spray to relieve chest pain, and a total of 3 doses can be given in 5-minute intervals if necessary. (The mechanism of pain relief with nitroglycerin is unclear; i.e., it is often attributed to its vasodilator effects, but other vasodilators do not relieve ischemic chest pain.) If the pain recurs, a nitroglycerin infusion can be started using the dosing regimen in Table 20.2. Nitroglycerin infusions can also benefit cases of ACS that are accompanied by hypertension or decompensated heart failure.

CONCERNS: Nitroglycerin is not advised in patients with right ventricular infarction (because the venodilator effects of nitroglycerin can be counterproductive in this condition), severe aortic stenosis, or in patients who have received a phosphodiesterase inhibitor for erectile dysfunction within the past 24–48 hours (because of the risk of hypotension) (2,5,6). (Complications of nitroglycerin infusions are described in Chapter 18.)

Morphine

Morphine is the drug of choice for chest pain that is refractory to nitroglycerin. Morphine also has a sedating effect, but the addition of a benzodiazepine (e.g., midazolam, 0.01 mg/kg IV) is sometimes needed to relieve anxiety.

CONCERNS: The complications of opioids are described in Chapter 6. Morphine (and fentanyl) can also impair the antiplatelet effects of the P2Y12 inhibitors used in ACS (see later), but the clinical significance of this effect is unproven (2).

β-Receptor Antagonists

β-blockers have several actions that reduce myocardial O_2 demand, and are usually started in the first 24 hours after diagnosis of ACS. The agent most often studied in ACS is *metoprolol*, and the dosing recommendations are shown in Table 20.3. Intravenous metoprolol can be useful in patients with troublesome tachycardia or hypertension, or prior to PCI in patients with STEMI (30).

When to Avoid

Traditional contraindications to β-blockers include bradycardia, hypotension, and high-degree

AV block. Other caveats for ACS are as follows:

- Early treatment with β-blockers is not advised in ACS associated with acute heart failure (2,5,6), but after the condition is stabilized, β-blockers are recommended for long-term therapy of heart failure with reduced ejection fraction (see Table 18.5).
- When the systolic BP is <120 mm Hg, β-blockers increase the risk of cardiogenic shock (31).
- β-blockers are contraindicated in cocaine-induced myocardial ischemia because they promote unopposed α-adrenergic vasoconstriction (32).

ANTITHROMBOTIC MEASURES

Antithrombotic measures are used to prevent growth of an existing thrombus (prior to reperfusion), and to reduce the risk of recurrence (after reperfusion).

Antiplatelet Agents

Antiplatelet therapy has a more important role in coronary thrombosis than in other thrombotic conditions (probably because of the vascular manipulation involved in coronary angioplasty). The antiplatelet agents and recommended dosing regimens for ACS are shown in Table 20.4.

Aspirin

Aspirin produces irreversible inhibition of platelet aggregation by inhibiting thromboxane production. It has a proven survival benefit in ACS (1 life saved per 42 treated) (33), and is given to all patients with suspected ischemia, as soon as possible after first patient contact. The initial dose is given as a chewable (non-enteric coated) formulation, and is intended for buccal absorption (i.e., should not be swallowed). Rectal aspirin is advised when oral delivery is not possible. Aspirin is continued indefinitely in cases of ACS, often together with a second antiplatelet agent (see next). For patients with an aspirin allergy, clopidogrel is a suitable alternative.

TABLE 20.4 Antiplatelet Measures for Acute Coronary Thrombosis

Agent	Dosing Regimen
Aspirin	162–325 mg (in chewable form) at first patient contact, then daily dose of 81 mg (for dual anti-platelet R_x) or 325 mg. If aspirin allergy, use clopidogrel.
P2Y12 Inhibitors	
Clopidogrel (Plavix)	PO: 300–600 mg initially, then 75 mg daily
Ticagrelor (Brilinta)	PO: 180 mg initially, then 90 mg twice daily
Prasugrel (Effient)	PO: 60 mg initially, then 10 mg daily
GP IIb-IIIa	IV: 180 μg/kg bolus (max 22.6 mg), then 2 μg/kg/min (max 15 mg/hr) with second bolus 10 min

Inhibitors		after first. Reduce infusion rate by 50% for Cr CL <50 mL/min.
Eptifibatide (Integrilin)		IV: 25 µg/kg (bolus), then 0.15 µg/kg/min. Reduce infusion rate by 50% for CrCL<30 mL/min.
Tirofiban (Aggrastat)		

From the clinical practice guidelines in References 2, 5, and 6.

P2Y12 Inhibitors

P2Y12 inhibitors block the platelet surface receptors that mediate ADP-induced platelet aggregation. The drugs available for use include clopidogrel, prasugrel and ticagrelor, and their dosing regimens are shown in Table 20.4. (Cangrelor is an intravenous P2Y12 inhibitor that is rarely used, and is not included here.) *A P2Y12 inhibitor should be added to aspirin (dual antiplatelet therapy) for all patients with ACS, and the first dose should be given prior to PCI* (2,5,6).

CLOPIDOGREL (PLAVIX): Clopidogrel is a prodrug that is converted to its active form in the liver, and this results in variable antiplatelet effect. Activation in the liver is also blocked by proton pump inhibitors (34), but the clinical significance of this is questioned. Once activated, clopidogrel causes irreversible platelet inhibition, and the drug should be stopped at least 5 days before a major surgical procedure. This prolonged effect is also reason to avoid clopidogrel if coronary bypass surgery is imminent.

TICAGRELOR (BRILINTA): Ticagrelor is a direct-acting drug that provides more potent and more consistent P2Y12 inhibition than clopidogrel, and it is superior to clopidogrel for improving outcomes (mortality rate, reinfarction) in patients with ACS (35). There is, however, an increased risk of intracranial hemorrhage (ICH) with ticagrelor (35), and the drug is contraindicated in patients with a prior ICH.

PRASUGREL (EFFIENT): Prasugrel is a prodrug like clopidogrel, but is more effective as an antiplatelet agent, and is superior to clopidogrel for preventing stent thrombosis and recurrent ischemia (36). However, the bleeding risk is higher with prasugrel (36), and the drug is contraindicated in patients with recent stroke or TIA (2,5,6).

Glycoprotein Receptor Antagonists

When platelets are activated, specialized glycoprotein receptors on the platelet surface (named IIb and IIIa) change their configuration and begin to bind fibrinogen. This allows fibrinogen molecules to form bridges between adjacent platelets, which promotes platelet aggregation. Glycoprotein receptor antagonists (also called *IIb/IIIa inhibitors*) block fibrinogen binding to activated platelets and inhibit platelet aggregation. These drugs are the most potent antiplatelet agents available, and are sometimes referred to as the *superaspirins*.

The IIb/IIIa inhibitors include *eptifibatide* (Integrilin), and *tirofiban* (Aggrastat). They are given by intravenous infusion, using the dosing regimens in Table 20.4. These drugs are used in high-risk patients who receive emergent PCI, and are given just before or at the start of the procedure. They are managed primarily by invasive cardiologists, and are not described further here.

Anticoagulant Therapy

Anticoagulation with heparin is recommended for all patients with ACS, and is started at the time of diagnosis. The options are as follows:

. Unfractionated heparin is preferred for patients who receive emergent PCI or thrombolysis. The recommended dose is 60 Units as an IV bolus (maximum 4,000 Units), followed by an infusion of 12 Units/kg/hr (maximum 1,000 Units/hr) initially, titrated to achieve an activated PTT(aPTT) of 1.5–2 times control (3,5). This is typically continued for 24–48 hrs.

. Low-molecular-weight heparin can be used when PCI is non-emergent or deferred. The dosing regimen for *enoxaparin* is 30 mg as an IV bolus, followed in 15 minutes by 1 mg/kg by subcutaneous injection every 12 hrs (3,5). (The bolus dose is not recommended in patients older than 75 years of age.) The dose is reduced by 50% (e.g., 1 mg/kg every 24 hours) when the creatinine clearance is <30 mL/min.

. For patients with a history of heparin-induced thrombocytopenia (see Chapter 19), the following alternative regimens are available (6):

a. For PCI: *Bivalirudin* (a direct thrombin inhibitor), 0.75 mg/kg as IV bolus, followed by an infusion of 1.75 mg/kg/hr. Reduce infusion rate to 1 mg/kg/hr for creatinine clearance <30 mL/min. Discontinue after successful PCI.

b. For Thrombolytic Therapy: *Fondaparinux* (factor Xa inhibitor), 2.5 mg as IV bolus, and on following day, start 2.5 mg by subcutaneous injection daily until cardiac catheterization. Contraindicated if creatinine clearance <30 mL/min.

LONG-TERM THERAPIES

The treatments summarized here are instituted when patients are clinically stable (e.g., after emergent PCI), and are continued indefinitely, as tolerated.

. High-intensity statin treatment with *atorvastatin*, 80 mg daily, has been shown to reduce the risk of major cardiovascular events following ACS (37), and is recommended for all patients with ACS (5,6).

. Inhibition of the renin-angiotensin-aldosterone (RAA) system is recommended for all patients with ACS, especially patients with hypertension, anterior STEMI, or heart failure associated with reduced ejection fraction (6). At the present time, the favored drug for this purpose is *sacubitril/valsartan* (Entresto) (38), a combination drug with an angiotensin receptor blocker, and a drug (sacubitril) that inhibits neprilysin (an enzyme that degrades natriuretic peptides). The dosing of this drug is shown in Table 18.5. RAA inhibitors are contraindicated in patients with a history of angioedema (from any source), and they should be used cautiously in patients with renal impairment who are prone to hyperkalemia.

. Long-term treatment with β-blockers is recommended for all patients with ACS, especially in cases of heart failure associated with reduced ejection fraction (see Chapter 18). Metoprolol is the β-blocker most often used in this setting, and the dosing recommendations are shown in Table 20.3.

ACUTE AORTIC DISSECTION

At the outset, the following points about acute dissection of the ascending aorta deserve emphasis:

- The clinical presentation can mimic, or include, ACS.
- This condition is a surgical emergency, and about 60% of patients will perish without prompt surgical intervention (39).
- One of every three cases is missed (40).

Pathophysiology

Aortic dissection occurs when a tear in the aortic intima allows blood to dissect between the intimal and medial layers of the aortic wall, creating a false lumen. This process can be the result of atherosclerotic damage, or accelerated degradation of the aortic wall from a genetic disorder (e.g., Marfan's syndrome). The dissection can originate in the ascending or descending aorta, and can propagate in antegrade and retrograde directions. When a dissection involves the ascending aorta between the aortic valve and the brachiocephalic artery (type A dissection), retrograde propagation can cause aortic valve insufficiency, coronary artery occlusion, and pericardial tamponade,while antegrade propagation can lead to neurologic deficits from occlusion of the aortic arch vessels.

Clinical Presentation

The most common complaint is the abrupt onset of sharp chest pain, which may be described as "ripping" or "tearing", and can be substernal (ascending aortic dissection) or in the back (descending aortic dissection). Only 5% of patients with acute aortic dissection are pain-free (41). Most importantly, *the chest pain can subside spontaneously for hours to days* (42,43), and the return of chest pain is often a sign of impending aortic rupture. *The spontaneous resolution of chest pain is an important source of missed diagnoses*.

Clinical Findings

The most frequent clinical findings are hypertension (50% of patients) and aortic insufficiency (50% of patients) (42,44). Unequal pulses in the upper extremities (from obstruction of the left subclavian artery) is a classic but infrequent finding (15% of cases) (42). The chest x-ray can show mediastinal widening (60% of cases) (42), but normal chest x-rays are reported in up to 20% of cases (41). The ECG can show ischemic changes (15% of cases) but it is normal in 30% of cases (41). Because of the limited sensitivity of clinical findings, additional imaging studies are required for the diagnosis.

Diagnostic Imaging

The diagnosis of aortic dissection requires one of four imaging modalities (43): magnetic resonance imaging (MRI) (sensitivity and specificity 98%), transesophageal echocardiography (sensitivity 98%, specificity 77%), contrast enhanced computed tomography (sensitivity 94%, specificity 87%), and aortography (sensitivity 88%, specificity 94%). As indicated, *MRI is the* *most sensitive and specific imaging modality for the diagnosis of aortic dissection*, but CT angiography is also acceptable.

The CT image in Figure 20.4 shows a dissection of the ascending aorta. The small arrows point to the intimal flap that separates the dissecting blood in the wall of the aorta (false lumen) from blood in the true lumen of the aorta. The presence of this flap distinguishes an aortic dissection from a saccular aneurysm of the aorta.

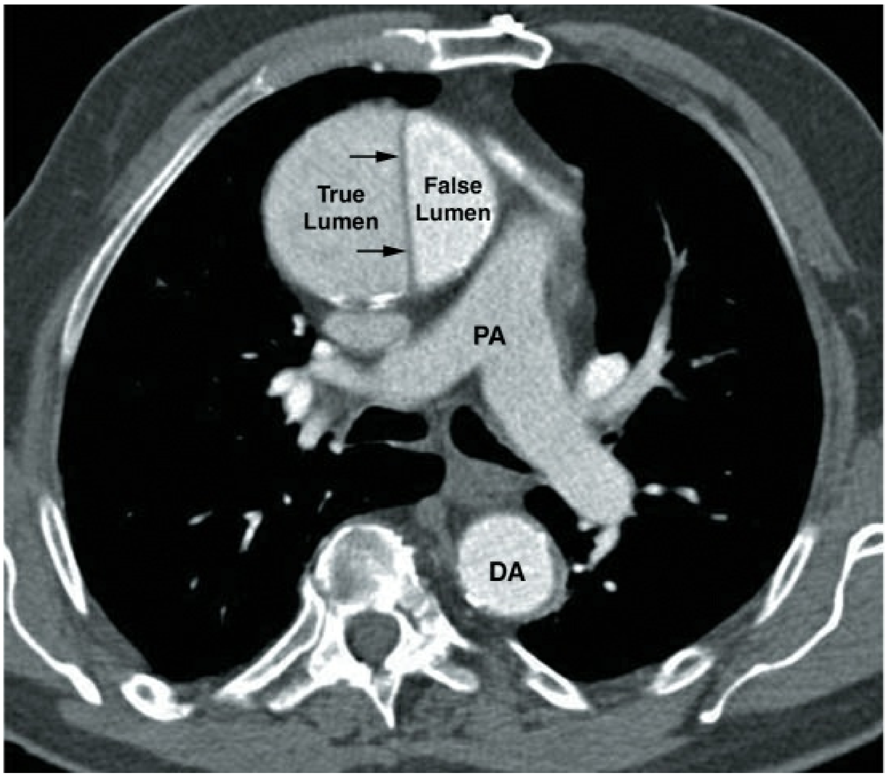

FIGURE 20.4 Contrast-enhanced CT image showing an acute dissection in the ascending aorta. The intimal flap that separates the true and false lumens (indicated by the small arrows) distinguishes an aortic dissection from a saccular aneurysm. PA = pulmonary artery, DA = descending aorta.

Management

The management goals in acute, type A aortic dissection include control of hypertension (to prevent aortic rupture), relief of pain (to assist in management of blood pressure), and prompt surgical intervention (the treatment of choice).

TABLE 20.5 Antihypertensive Therapy for Acute Aortic Dissection

Drug	Dosing Regimens and Comments
Esmolol	Dosing: 500 µg/kg as IV bolus, then infuse at 50 µg/kg/min and increase rate in increments of 25 µg/kg/min to desired BP, or to maximum of 200 µg/kg/min. Give a repeat bolus dose before each increment in infusion rate. Comment: Ultra rapid-acting β-blocker that can be rapidly titrated to achieve the desired BP. Not advised in the presence of acute heart failure.

Labetalol	Dosing: 20 mg IV over 2 min, then 20–40 mg IV every 10 min as needed, or infuse at 1–2 mg/min and titrate to desired BP. Max. cumulative dose is 300 mg. Comment: A combined α- and β-blocker that is used as monotherapy. Avoid use in acute heart failure.
Metoprolol	Dosing: 5mg as IV bolus and repeat in 5 min x 2, if needed. Continue with 5–10 mg IV every 4–6 hrs, as needed. Comment: Best suited for initial β-blockade when vasodilators are used.
Nicardipine	Dosing: Infuse at 5mg/hr, and increase by 2.5 mg/hr every 5 min as needed to maximum infusion of 15 mg/hr. Comment: Use in combination with a β-blocker.
Nitroprusside	Dosing: Infuse at 0.2 µg/kg/min and titrate upward every 5 min to the desired effect. Effective dose is typically 2–5 µg/kg/min, but avoid prolonged infusions at >3 µg/kg/min to reduce the risk of cyanide toxicity. Can add thiosulfate (500 mg) to the infusate (binds cyanide released by nitroprusside. Comment: Use in combination with a β-blocker. Do not use in patients with hepatic or renal failure, or in the presence of coronary ischemia.

Dosing regimens are manufacturer's recommendations.

Antihypertensive Therapy

There is one major caveat for blood pressure control in aortic dissection; i.e., *the reduction in blood pressure should NOT be accompanied by tachycardia or an increase in cardiac output*, as these conditions will augment the shear forces that promote dissection. As a result, β-*blockers are preferred* for blood pressure control in aortic dissection, and vasodilators (e.g., nicardipine) should only be used in combination with β-blockers (39). The drug regimens that are used in aortic dissection are shown in Table 20.5. Esmolol is the preferred β-blocker because it is rapidly titratable. The goal of antihypertensive management in aortic dissection is a systolic pressure of 120 mm Hg, although pressures down to 90 mm Hg are usually tolerated (39).

Pain Relief

Pain can aggravate an acute aortic dissection by activating the sympathetic nervous system, which promotes an increase in heart rate, cardiac output, and blood pressure. Thus, pain relief with intravenous opioids is an early goal of management (39).

Surgical Intervention

Open surgical repair of the ascending aorta is the standard of care in patients with acute, type A aortic dissection, and timely intervention is critical, because the mortality rate increases 1–2% per hour after the onset of symptoms (39). If the patient is at a hospital that does not provide acute cardiac surgery, then transfer to another hospital for surgical intervention is imperative. Surgical repair can reduce the mortality rate to as low as 10% (30).

A FINAL WORD

The O_2 Supply-Demand Dogma

The traditional notion that myocardial infarctions are the result of an imbalance between O_2 supply and O_2 demand is misleading, because it implies that conditions like anemia and hypoxemia can cause myocardial infarctions. However, *myocardial infarctions are caused by a blood clot that obstructs a coronary artery, not by conditions that cause a global decrease in myocardial O_2 delivery*.

Ever wonder why acute MIs do not occur in patients with progressive circulatory shock (where O_2 delivery is severely impaired)? Now you have the answer.

References

1. Tsao CW, Aday AW, Almarzoog ZI, et al. Heart disease and stroke statistics—2023 update: a report from the American Heart Association. Circulation 2023; 147:e93–e621.

Clinical Practice Guidelines

2. Byrne RA, Rosello X, Coughlan JJ, et al. 2023 ESC Guidelines for the management of acute coronary syndromes. Developed by the task force on the management of acute coronary syndromes by the European Society of Cardiology (ESC). Eur Heart J 2023; 44:3720–3826.
3. Byrne RA, Rosello X, Coughlan JJ, et al. 2023 ESC Guidelines for the management of acute coronary syndromes. Supplementary data. Eur Heart J 2023; 00:1–52.
4. Gulati M, Levy PD, Mukherjee D, et al. AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR guideline for the evaluation and diagnosis of chest pain: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 2021; 144: e368–e454.
5. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC guideline for the management of patients with non–ST-elevation acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol 2014; 64:e139–228.
6. O'Gara PT, Kushner FG, Ascheim DD, et al. 2013 ACCF/AHA guideline for the management of ST-elevation myocardial infarction: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. J Am Coll Cardiol 2013; 61:e78 –140.
7. Lawton JS, Tamis-Holland JE, Bangalore S, et al. 2021 ACC/AHA/SCAI guideline for coronary artery revascularization: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. J Am Coll Cardiol 2022; 79:e21–e129.

Coronary Thrombosis

8. Van der Wal AC, Becker AE, van der Loos CM, Das PK. Site of intimal rupture or erosion of thrombosed coronary atherosclerotic plaques is characterized by an inflammatory process irrespective of the dominant plaque morphology. Circulation 1994; 89:36–44.
9. Malek AM, Alper SL, Izumo S. Hemodynamic shear stress and its role in atherosclerosis. JAMA 1999; 282:2035–2042.
10. Swirski FK, Nahrendorf M. Leukocyte behavior in atherosclerosis, myocardial infarction, and heart failure. Science 2013; 339:161–166.
11. Daugherty A, Dunn JL, Rateri DL, Heinecke JW. Myeloperoxidase: a catalyst for lipoprotein oxidation, is expressed in human atherosclerotic lesions. J Clin Invest 1994; 94:437–444.
12. Teng N, Maghzal GJ, Talib J, et al. The roles of myeloperoxidase in coronary artery disease and its potential implication in plaque rupture. Redox Report 2017; 22:51–73.
13. Ndrepepa G. Myeloperoxidase—A bridge linking inflammation and oxidative stress with cardiovascular disease. Clin Chim Acta 2019; 493:36–51.
14. Swamy N. Esophageal spasm: clinical and manometric response to nitroglycerine and long acting nitrates. Gastroenterology 1977; 72:23–27.
15. Thygesen K, Mair J, Giannitsis E, et al. Study Group on Biomarkers in Cardiology of the ESC Working Group on Acute Cardiac Care. How to use high-sensitivity cardiac troponins in acute cardiac care. Eur Heart J 2012; 33:2252–2257.

Reperfusion Strategies

16. Cannon CP, Gibson CM, Lambrew CT, et al. Relationship of symptom onset to balloon time and door-to-balloon time with mortality in patients undergoing angioplasty for acute myocardial infarction. JAMA 2000; 283:2941–2947.
17. Concannon TW, Nelson J, Goetz J, Griffith JL. A percutaneous coronary intervention lab in every hospital. Circ Cardiovasc Qual Outcomes 2012; 5:14–29.
18. Wang TY, Peterson ED, Ou FS, et al. Door-to-balloon times for patients with ST-segment elevation myocardial infarction requiring interhospital transfer for primary percutaneous coronary intervention: a report from the national cardiovascular data

registry. Am Heart J 2011; 161:76–83.
19. Keeley EC, Boura JA, Grines CL. Primary angioplasty versus intravenous thrombolytic therapy for acute myocardial infarction: a quantitative review of 23 randomized trials. Lancet 2003; 361:13–20.
20. Fibrinolytic Therapy Trialists Collaborative Group. Indications for fibrinolytic therapy in suspected acute myocardial infarction: collaborative overview of early mortality and major morbidity results from all randomized trials of more than 1000 patients. Lancet 1994; 343:311–322.
21. Smalling RW, Bode C, Kalbfleisch J, et al. More rapid, complete, and stable coronary thrombolysis with bolus administration of reteplase compared with alteplase infusion in acute myocardial infarction. Circulation 1995; 91:2725–2732.
22. GUSTO-III Investigators. An international, multicenter, randomized comparison of reteplase with alteplase for acute myocardial infarction. N Engl J Med 1997; 337:1118–1123.
23. Llevadot J, Giugliano RP, Antman EM. Bolus fibrinolytic therapy in acute myocardial infarction. JAMA 2001; 286:442–449.
24. Assessment of the Safety and Efficacy of a New Thrombolytic (ASSENT-2) Investigators. Single-bolus tenecteplase compared with front-loaded alteplase in acute myocardial infarction. Lancet 1999; 354:716–722.
25. Berkowitz SD, Granger CB, Pieper KS, et al. Incidence and predictors of bleeding after contemporary thrombolytic therapy for myocardial infarction. Circulation 1997; 95:2508–2516.
26. Young GP, Hoffman JR. Thrombolytic therapy. Emerg Med Clin 1995; 13:735–759.

Cardioprotective Measures

27. Hofmann R, James SK, Jernberg T, et al. Oxygen therapy in suspected acute myocardial infarction. N Engl J Med 2017; 377:1240–1249.
28. Farquhar H, Weatherall M, Wijesinghe M, et al. Systematic review of studies of the effects of hyperoxia on coronary blood flow. Am Heart J 2009; 158:371–377.
29. Bulkley GB. Reactive oxygen metabolites and reperfusion injury: aberrant triggering of reticuloendothelial function. Lancet 1994; 344:934–936.
30. Roolvink V, Ibáñez B, Ottervanger JP, et al. Early intravenous beta-blockers in patients with ST-segment elevation myocardial infarction before primary percutaneous coronary intervention. J Am Coll Cardiol 2016; 67:2705–2715.
31. Chen ZM, Pan HC, Chen YP, et al. Early intravenous then oral metoprolol in 45,852 patients with acute myocardial infarction: randomized placebo-controlled trial. Lancet 2005; 366:1622–1632.
32. McCord J, Kneid H, Hollander JE, et al. Management of cocaine-associated chest pain and myocardial infarction. A scientific statement from the American Heart Association Acute Cardiac Care Committee and the Council on Clinical Cardiology. Circulation 2008; 117:1897–1907.

Antithrombotic Measures

33. ISIS-2 Collaborative Group. Randomized trial of intravenous streptokinase, oral aspirin, both, or neither among 17,187 cases of suspected acute myocardial infarction: ISIS-2. Lancet 1988; 2:349–360.
34. Bhatt DL, Cryer BL, Contant CF, et al. Clopidogrel with or without omeprazole in coronary artery disease. N Engl J Med. 2010; 363:1909–1917.
35. Wallentin L, Becker RC, Budjai A, et al; for the PLATO Investigators. Ticagrelor versus clopidogrel in patients with acute coronary syndromes. N Engl J Med 2009; 361:1045–1057.
36. Wiviott SD, Braunwald E, McCabe CH, et al. Prasugrel versus clopidogrel in patients with acute coronary syndromes. N Engl J Med 2007; 357:2001–2015.

Long-Term Therapies

37. Cannon C, Braunwald E, McCabe CH, et al. Intensive versus moderate lipid lowering with statins after acute coronary syndromes. N Engl J Med 2004; 350:1495–1504.
38. Velazquez EJ, Morrow DA, DeVore AD, et al. Angiotensin-neprilysin inhibition in acute decompensated heart failure. N Engl J Med 2019; 380:539–548.

Aortic Dissection

39. Malaisrie SC, Szeto WY, Halas M, et al. 2021 The American Association for Thoracic Surgery expert consensus document: Surgical treatment of acute type A aortic dissection. J Thorac Cardiovasc Surg 2021; 162:735–758.
40. Lovatt S, Wong CW, Schwarz K, et al. Misdiagnosis of aortic dissection: A systematic review of the literature. Am J Emerg Med 2022; 53:16–22.
41. Tsai TT, Nienaber CA, Eagle KA. Acute aortic syndromes. Circulation 2005; 112:3802–3813.
42. Knaut AL, Cleveland JC. Aortic emergencies. Emerg Med Clin N Am 2003; 21:817–845.
43. Zegel HG, Chmielewski S, Freiman DB. The imaging evaluation of thoracic aortic dissection. Appl Radiol 1995; (June):15–25.
44. Khan IA, Nair CK. Clinical, diagnostic, and management perspectives of aortic dissection. Chest 2002; 122:311–328.

Chapter 21

Cardiac Arrest

When we all think alike, then no one is thinking.

Walter Lippmann (*a*)

In 1960, a report appeared in the *Journal of the American Medical Association* that would eventually change the way we approach the dying process. The report was titled "Closed Chest Cardiac Massage" (1), and it included 5 cases of cardiorespiratory arrest that were successfully managed with chest compressions, electrical shocks, and assisted ventilation. That same year (and without any validation studies), the American Heart Association started a program to educate physicians about closed-chest cardiac resuscitation, which eventually grew to become *cardiopulmonary resuscitation* (CPR). Since its inception, CPR has become a universally mandated practice that is withheld only on request, and requires certification every 2 years to ensure the competence of practitioners. And all this for a treatment modality that fails in most cases, as shown in Table 21.1 (2).

This chapter describes the practical aspects of CPR, and the management that follows a successful resuscitation effort. The recommendations in this chapter are taken from the most recent and most relevant guidelines and instruction manuals on CPR, which are listed in the bibliography at the end of the chapter (3–8).

TABLE 21.1 Cardiac Arrest Data for the United States, 2021

	Out-of-Hospital Cardiac Arrests	In-Hospital Cardiac Arrests
Number	143,018	8,619
Vtach/Vfib	17%	13%
PEA/Asystole	75%	78%
Survived to Hospital Discharge	9%	19%
Survived with Good Functional Status	7%	1%

Data from Reference 2.

BASIC LIFE SUPPORT

The elements of basic life support (BLS) are chest compressions, and periodic lung inflations through a patent upper airway. The priority for these elements is indicated by the popular mnemonic *CAB* (Circulation, Airway, and Breathing).

Time Dependence

One of the major limiting factors in the success of CPR is the narrow time window between the cessation of blood flow and irreversible cell death. The total O_2 content of the adult human body is about one liter, and with a normal O_2 consumption of about 250 mL/min at rest, this leaves only about 4 minutes after the cessation of blood flow until the O_2 content of the body is completely depleted. Thus, *following the cessation of blood flow from a cardiac arrest, anoxic cell death is expected in just 4–5 minutes*. (One exception to this is cold-water drowning, where the cold temperatures will reduce O_2 consumption and prolong the time for anoxic cell injury.) This emphasizes the "need for speed" in initiating CPR.

TABLE 21.2	The Essential Elements of Basic Life Support
1. Chest compressions should begin within 10 seconds of detecting the absence of pulses.	
2. Each chest compression should depress the lower third of the sternum by 2–2.4 inches (5–6 cm), and the chest should be allowed to recoil completely before the next compression.	
3. The rate of compressions should be 100–120/minute.	
4. After 30 chest compressions, 2 lung inflations are delivered (with a bag-mask device) without interrupting the compressions, and this (30:2) cycle is repeated until the patient is intubated.	

5. Following intubation, lung inflations are delivered every 6 seconds (10/min) without interrupting chest compressions. However, simultaneous chest compressions and lung inflations should be avoided.
6. Chest compressions should be continued without interruption until a defibrillator is attached to the patient.
7. Each person performing chest compressions should be relieved after 2 minutes, if possible.

From References 3–5, 7.

Overview

The trigger for CPR is the absence of pulses in an unresponsive patient who has minimal or no spontaneous breathing efforts. Chest compressions should begin within 10 seconds of this trigger, using the depth and frequency of compressions shown in Table 21.2. Two lung inflations should be delivered (with a bag-mask device) after every 30 chest compressions, without interrupting the compressions (3–5). After an endotracheal (ET) tube is placed, lung inflations are delivered at 6-second intervals (10/min) without interrupting chest compressions. However, simultaneous chest compression and lung inflation will limit the inflation volume, and should be avoided (4). Finally, an ECG monitor/defibrillator should be attached to the patient as soon as possible.

Chest Compressions

The backbone of BLS is high-quality chest compressions with minimal interruptions. Observational studies have shown that interruptions in chest compressions can account for as much as 50% of the total resuscitation time (9), and prolonged interruptions could contribute to poor outcomes. Current recommendations are to continue chest compressions without interruptions until a defibrillator is attached to the patient. (Thereafter, compressions can be paused to deliver electric shocks, and to check the cardiac rhythm, as described later.)

High-Quality Compressions

High-quality chest compressions are defined as having a depth of 2–2.4 inches (5–6 cm) and a rate of 100 to 120/min, with full chest recoil allowed between compressions (3–5). Estimating the depth of compressions is difficult, especially with a target range of only 0.4 inches. There are devices used for CPR training that measure the depth of compression and provide audio feedback (e.g., using the sometimes irritating prompt, “push harder”), but they are not available for clinical use. Fatigue can influence the depth of compressions, and each person performing compressions should be replaced after 2 minutes, if possible (4,5).

Lung Inflations

Lung inflations are delivered by compressing a self-inflating ventilation bag (e.g., Ambu Respirator) that fills with oxygen. The recommended volume for each lung inflation is 500–600 mL, or whatever produces a visible rise in the chest wall (4). However, inflation volumes are not monitored during “bagged ventilation”, and large inflation volumes are common during CPR (10), which can be damaging for the lungs (similar to “ventilator-induced lung injury”, which is described in Chapter 24). Closer adherence to the recommended inflation volumes is possible if the volume capacity of the inflation bags in known. For example, if the inflation bag has a volume capacity of 1 liter, compressing the bag until it is about half full will deliver a volume of about 500 mL. (The volume capacity of most adult ventilation bags is 1 to 2 liters.) An alternative approach is to use one hand to compress the ventilation bag, which delivers a volume of about 600 to 700 mL (personal observation), and is unlikely to produce troublesome lung inflations.

Rapid Lung Inflations

Rapid lung inflations are common during CPR (11,12), and average rates of 30 inflations/min (3 times the recommended rate) have been reported (12). Rapid breathing is problematic because there is insufficient time for the lungs to empty, which leads to progressive hyperinflation and positive end-expiratory pressure (PEEP). The PEEP that is produced by rapid breathing is called "intrinsic PEEP", and is described in detail in Chapter 23. The increase in mean intrathoracic pressure from PEEP has two deleterious effects: 1) it reduces venous return to the heart, which limits the ability of chest compressions to augment cardiac output, and 2) it reduces coronary perfusion pressure (11), which is an important determinant of outcome in cardiac arrest. For these reasons, avoiding rapid inflation rates will increase the chances for a favorable outcome with CPR.

Vascular Access

Vascular access should be established without interrupting chest compressions. Peripheral venous access is considered suitable (4), but intraosseous access is often favored because it is easy to establish without interrupting chest compressions, and is more secure than a peripheral vein catheter. (See Chapter 1 for a description of intraosseous access.)

ADVANCED LIFE SUPPORT

Advanced cardiovascular life support (ACLS) includes electrical cardioversion (defibrillation) and the use of drugs that support the resuscitation attempt. This approach divides the management of cardiac arrest into two pathways: one for the management of ventricular fibrillation (VF) and pulseless ventricular tachycardia (VT), and the other for pulseless electrical activity (PEA) and asystole.

VF and Pulseless VT

The principal intervention for VF and pulseless VT is electrical "defibrillation", and hence these arrhythmias are referred to as "shockable rhythms".

Defibrillation

Electrical cardioversion that uses asynchronous shocks (i.e., not timed to the QRS complex) is called *defibrillation*, and it is one of the few interventions that improves survival in cardiac arrest victims. However, the survival benefit is time-dependent. This is shown in Figure 21.1, which is from a study of the relationship between survival and the time (from collapse) to defibrillation in cardiac arrests associated with VF or VT (13). Note that 40% of patients survived when the first shock was delivered 5 minutes after the arrest, while fewer than 10% of patients survived if defibrillation was delayed until 20 minutes after the arrest.

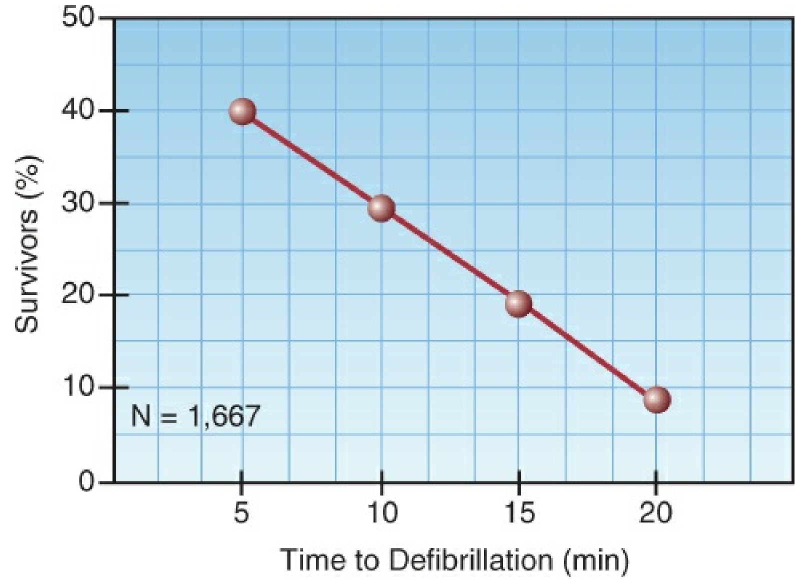

FIGURE 21.1 Relationship between survival and time (from collapse) to first defibrillator shock in 1,667 out-of-hospital cardiac arrests associated with VF or pulseless VT. Data from Reference 13.

IMPULSE ENERGY: Modern defibrillators deliver biphasic shocks (which convert VF and VT at lower energy levels than monophasic shocks), and the effective energy level can vary from as low as 100 J to as high as 360 J, with higher energy levels needed for resistant or recurrent episodes of VF/VT (14). Automated external defibrillators (AEDs) usually deliver fixed-energy shocks, while other defibrillators either have a pre-selected energy level for shocks, or allow the user to select the desired energy level.

Management Algorithm

The ACLS algorithm for cardiac arrest in adults is shown in Figure 21.2 (8), and the left side of the flow diagram indicates the management when the initial rhythm is VF or VT. The major features of the management are summarized below.

- The management includes a series of 3 defibrillation attempts, if needed. The initial shock is usually 120–200 J for biphasic shocks (or is manufacturer-recommended). If this is ineffective, a higher energy level can be used for subsequent shocks, if allowed by the defibrillator.
- Chest compressions are paused when the defibrillation shock is delivered, and are resumed immediately thereafter. At least 2 minutes of uninterrupted chest compressions are recommended after defibrillation before checking the post-shock rhythm.
- If a second defibrillation is needed, *epinephrine* is started using a bolus dose of 1 mg (intravenous or intraosseous) every 3–5 minutes for the duration of the resuscitation effort.
- If a third defibrillation is needed, *amiodarone* is administered as a bolus dose of 300 mg (IV or IO), which can be followed by a second dose of 150 mg, if needed. If amiodarone is not available, *lidocaine* can be given in a initial dose of 1–1.5 mg/kg (IV or IO), followed by 0.5–0.75 mg/kg every 5–10 min as needed, to a maximum dose of 3 mg/kg (8).

SHOCK-RESISTANT CASES: Failure of three defibrillation attempts to convert VT and VF has a very poor prognosis, with a satisfactory outcome in only 5% of cases (15). Improved outcomes have been reported in shock-resistant VF/VT that is managed with emergency extracorporeal life support (ECMO) (16), and this approach is certainly a consideration if ECMO is available on a 24-hour basis.

Asystole / PEA

The management of cardiac arrest associated with pulseless electrical activity (PEA) and ventricular asystole is shown on the right half of the ACLS algorithm in Figure 21.2. The major intervention is vasopressor therapy with epinephrine using the same dosing regimen used for VF and pulseless VT. Defibrillation is not attempted unless the cardiac rhythm changes to VF or VT.

Reversible Causes of PEA

PEA has potentially reversible causes, which can be identified by the letter "T": i.e., Tension pneumothorax, pericardial Tamponade, venous Thromboembolism, and Thrombotic occlusion of the coronary arteries. Although there is little time for a diagnostic workup during a cardiac arrest, point-of-care ultrasound can be a valuable aid in uncovering some of these conditions (see later).

Epinephrine

Epinephrine is the only circulatory-support drug that is used for CPR. Although it has been shown to increase the rate of return of spontaneous circulation (17), the impact on survival is not clear. Most studies show no improvement in survival associated with epinephrine (17,18), but there is one study that shows an increase in survival at 30 days (19), although many of the survivors were not mentally intact.

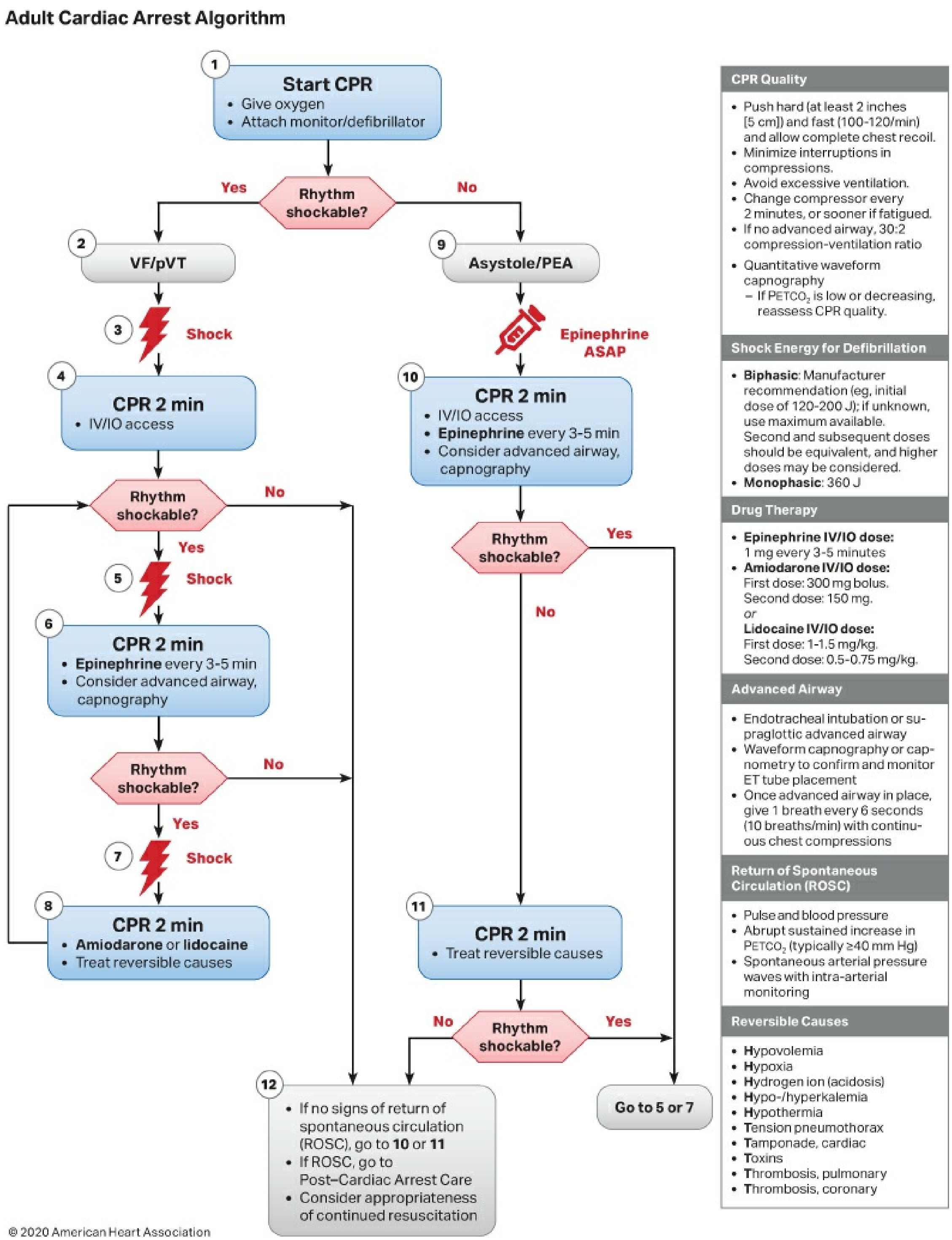

FIGURE 21.2 The American Heart Association algorithm for ACLS. Reprinted with permission Circulation. 2020; 142:S366–S468 © 2020 American Heart Association, Inc.

Actions

The benefit derived from epinephrine is the systemic vasoconstriction, which can increase flow in both the cerebral and coronary circulations. The increase in coronary blood flow is due to an increase in coronary perfusion pressure (the difference between aortic and right atrial pressures between chest compressions). This is demonstrated in Figure 21.3 (20). In this case, there is a 30% increase in coronary perfusion pressure following IV epinephrine, and the effect lasts at least 3 minutes (the recommended time interval between epinephrine doses).

The disadvantage with epinephrine is the β-receptor-mediated cardiac stimulation, which can erase the benefit of increased coronary perfusion, and has also been implicated in post-resuscitation heart failure (20).

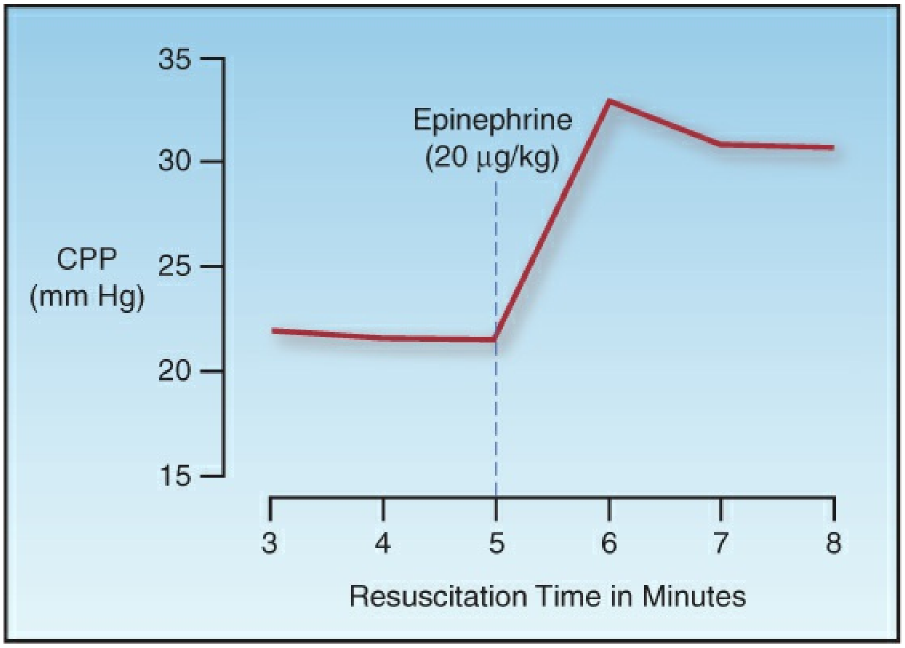

FIGURE 21.3 The effect of intravenous epinephrine on coronary perfusion pressure (CPP) during resuscitation of cardiac arrest with VF/pulseless VT. Data from Reference 15.

RESUSCITATION MONITORING

Monitoring for the return of spontaneous circulation (ROSC) has traditionally been limited to manual palpation for pulses, but this practice has a low sensitivity (21), and often requires interruption of chest compressions for extended periods of time (22). Ultrasound for carotid artery pulsations can speed the process of pulse detection (23), but also requires interruption of chest compressions. The end-tidal PCO_2 provides a more reliable (and physiological) evaluation of the circulation, and can be used to predict the likelihood of ROSC.

End-Tidal PCO_2

(The end-tidal PCO_2 measurement is described in detail in Chapter 7: see Figure 7.6.) The PCO_2 in exhaled gas at the end of expiration (end-tidal PCO_2) is a relection of the balance between ventilation and perfusion in the lungs. The end-tidal PCO_2 ($ETCO_2$) varies directly with changes in cardiac output relative to ventilation; i.e., *when alveolar ventilation is constant, changes in end-tidal PCO_2 reflect proportional changes in cardiac output* (e.g., a 30% decrease in end-tidal PCO_2 indicates a 30% decrease in the cardiac output). The baseline end-tidal PCO_2 is normally equivalent to the arterial PCO_2 (i.e., about 40 mm Hg), but it is lower than the arterial PCO_2 in pulmonary conditions associated with increased physiologic dead space (i.e., V/Q ratio >1), such as chronic obstructive lung disease.

Using the End-Tidal PCO_2

The $ETCO_2$ has been suggested for monitoring the quality of chest compressions; i.e., an increase in depth of compressions is accompanied by an increase in $ETCO_2$ (24). More importantly, serial measurements of $ETCO_2$ during CPR can be used to identify the likelihood of the return of spontaneous circulation (ROSC). This is demonstrated by the graph in Figure 21.4, which show the serial changes in $ETCO_2$ during CPR in relation to ROSC (25). Patients who achieved ROSC showed a progressive increase in $ETCO_2$ during the 20-minute resuscitation period, whereas patients who did not achieve ROSC showed a progressive decline in $ETCO_2$. The $ETCO_2$ that separated responders from nonresponders in this study was 15 mm Hg after 20 minutes. Other studies have shown a discriminant $ETCO_2$ of 10 mm Hg for separating responders from nonresponders (26,27).

The sum of the evidence indicates that *a successful resuscitation is unlikely if the end tidal PCO_2 is not higher than 10–15 mm Hg after 20 minutes of CPR*. When the end-tidal PCO_2 remains above this level, continued resuscitation for as long as 1½ hours has been associated with a favorable outcome (28).

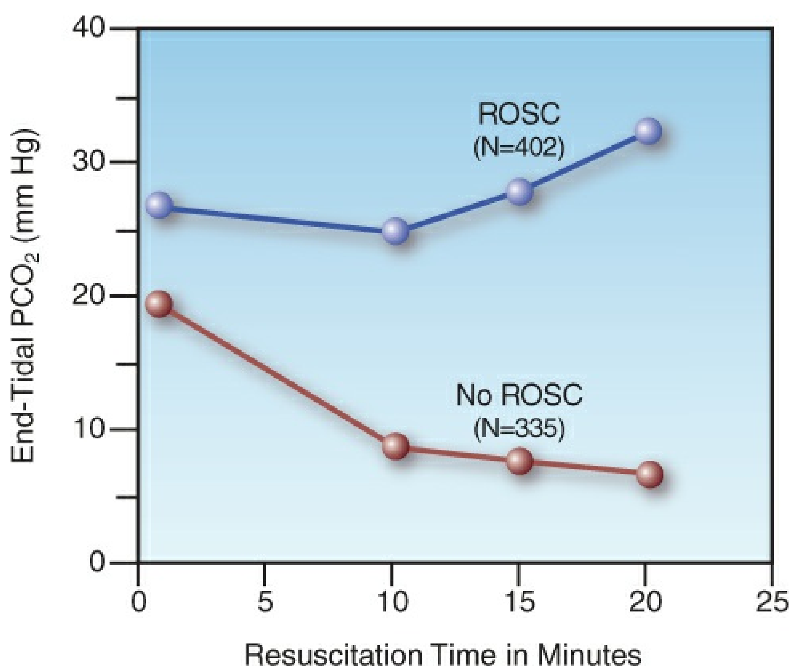

FIGURE 21.4 Serial changes in end tidal PCO_2 during CPR in relation to return of spontaneous circulation (ROSC) in 737 cases of out-of-hospital cardiac arrest. N indicates the number of patients in each group, and data points represent mean values. From Reference 25.

Ultrasound

Point-of-care ultrasound is invaluable for the detection of potentially reversible causes of cardiac arrest, such as tension pneumothorax or pericardial tamponade. Ultrasound has a high sensitivity (91%) and specificity (99%) for the detection of pneumothoraces in the supine position (29), and pericardial effusions are readily detected in the subcostal or parasternal long-axis views (see Figure 21.5) (30). The traditional signs of pericardial tamponade (i.e., diastolic collapse of the right ventricle) are unreliable during a cardiac arrest (due to low intracardiac pressures), so the presence of any pericardial effusion should prompt immediate pericardiocentesis.

Studies in trauma-associated cardiac arrest have shown that ultrasound evidence of cardiac standstill is 100% predictive of failure to survive (31). However, there is considerable disagreement in the recognition of cardiac standstill among physicians that perform point-of-care ultrasound (32); as a result, the clinical practice guidelines for cardiac arrest do not recommend the ultrasound detection of cardiac standstill as a reason to terminate CPR (4).

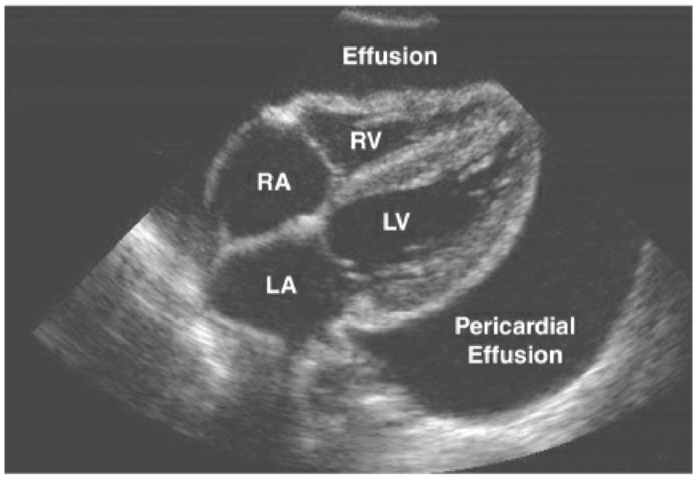

FIGURE 21.5 Ultrasound image (subcostal view) showing a large pericardial effusion, which appears as a broad anechoic band surrounding the heart. Image retouched from Loyola University Stritch School of Medicine, medical education curriculum in radiology, at www.stritch.luc.edu, accessed 11/18/2023.

POST-RESUSCITATION PERIOD

The immediate goal of CPR is a return of spontaneous circulation, but this does not ensure a satisfactory outcome. In fact, *about 70% of patients who survive the resuscitation do not survive the hospitalization* (33). This section describes the common problems encountered in the days following a successful resuscitation of cardiac arrest.

Post-Cardiac Arrest Syndrome

The return of spontaneous circulation is often followed by dysfunction in one or more major organs that can be progressive and ultimately fatal. This *post-cardiac arrest syndrome* is attributed to a combination of ischemic injury inflicted during the arrest, and a *reperfusion injury* from toxic substances released from ischemic tissues when blood flow is re-established (33,34). The principal features of this syndrome are summarized below:

- *Brain injury* is the most common manifestation of the post-cardiac arrest syndrome, and is *responsible for 23% to 68% of deaths following cardiac arrest* (33). Clinical manifestations include failure to awaken, myoclonus, and generalized seizures. The high prevalence of brain injury after cardiac arrest is attributed to a limited tolerance to ischemia, and a predisposition to oxidative reperfusion injury from reactive oxygen species (34).

- Post-arrest *cardiac dysfunction* is a combination of systolic and diastolic dysfunction that can progress to cardiogenic shock within hours after ROSC (33). The underlying problem is a type of reperfusion injury known as myocardial "stunning", which usually resolves within 72 hours (33).
- A systemic inflammatory response (i.e., fever, leukocytosis, etc.) is almost universal after cardiac arrest, and can result in widespread inflammatory injury with multiorgan failure and circulatory shock. (See Chapter 17 for more on inflammatory shock.)

Targeted Temperature Management

Increased body temperature is well-known for its ability to aggravate ischemic brain injury (35), and control of body temperature has been a major focus of efforts to limit neurologic injury in patients who survive a cardiac arrest.

A Brief History

Induced hypothermia (originally called *human refrigeration*) was introduced for the management of post-cardiac arrest victims in the 1950s (36), but was abandoned because of the risks associated with hypothermia. About a half century later (in 2002), two studies were published showing that mild hypothermia (32–34° C) improved the chances of neurologic recovery in survivors of cardiac arrest who remained comatose (37,38). This led to the immediate adoption of induced hypothermia at 32–34° C for comatose survivors of cardiac arrest. About 10 years later, a large multicenter study showed that a target temperature of 36° C was equivalent to 33° C for influencing outcomes (39), and also reduced the risk of hemodynamic compromise (40). After another 10 years, the same group of investigators showed that a target temperature of 37.5° C (normothermia) was equivalent to 33° C for influencing outcomes (41). This brings us to the current state of affairs, where *the goal of targeted temperature management is to prevent fever in comatose survivors of cardiac arrest*.

TABLE 21.3 Targeted Temperature Management

Feature	Recommendations†
Indication	Patients who do not regain consciousness after ROSC
Goal	Body Temp ≤37.5° C (≤99.5° F) for 72 hours
Monitoring	Continuous monitoring of core body temperature; e.g., with a thermistor-equipped bladder catheter
Treatment Plan	1. Patients with mild hypothermia (32°–36° C) after ROSC should not be actively rewarmed. 2. Maintain body temp at ≤37.5° C with acetaminophen and reduced room temp, if needed. 3. If body temp rises above 37.7° C (>99.9° F), start active cooling, and set target temp at 37.5° C (99.5° F). 4. Surface cooling is acceptable. 5. Body temp is kept at ≤37.5° C for 72 hours, unless the patient awakens.

†From the clinical practice guidelines in Reference 3.

The Method

The salient features of targeted temperature management (TTM) are outlined in Table 21.3 (3,41). Candidates for TTM include all patients who do not regain consciousness after return of spontaneous circulation (ROSC), and the goal is to maintain a body temperature ≤37.5° C (99.5° F) for 72 hours after ROSC, or until the patient awakens. Continuous monitoring of core body temperature is required, and is easily accomplished with thermistor-equipped bladder catheters. A temperature-controlled cooling device is used only if the body temperature rises above 37.7° C (99.9° F), and if this occurs, surface cooling should be adequate, with the target temperature set at 37.5° C.

Despite the recommendation for normothermic TTM in the most recent clinical practice guidelines (3), there is a reluctance to abandon hypothermic TTM, and the option to use a target temperature of 33° C is left unresolved. However, the induction of hypothermia introduces a number of undesirable factors, such as the risk of hypotension (from cold-induced diuresis and cardiac depression), bradycardia (common during hypothermia), and shivering, which often requires heavy sedation (which can delay the assessment of mental status) and occasionally requires neuromuscular paralysis (which is never desirable).

Other Concerns

In addition to preventing fever, there are other concerns for the management of patients in the early post-arrest period. These are summarized in Table 21.4.

TABLE 21.4 Other Concerns in Post-Arrest Management

Intervention	Comment
Oxygen Inhalation	Hyperoxia aggravates neurologic injury after cardiac arrest, so O_2 inhalation should be used only to correct hypoxemia (SaO_2 <90%).
Vasopressor R_x	Maintaining a mean arterial pressure higher than usual (e.g., ≥75 mm Hg) may benefit neurologic recovery.
Vasopressor Agent	Norepinephrine is preferred to epinephrine (see text for explanation).
Glycemic Control	Hyperglycemia aggravates neurologic injury after cardiac arrest, but avoid very tight glycemic control because hypoglycemia also aggravates neurologic injury.

Oxygen Inhalation

Hyperoxia can aggravate the neurologic injury following cardiac arrest (42), and oxygen inhalation should be used only to correct hypoxemia (arterial O_2 saturation <90%) (6). Reactive oxygen species are implicated in the reperfusion injury that follows return of spontaneous circulation (34), which provides another reason to limit O_2 breathing in the post-arrest period.

Managing Hypotension

Ischemic brain injury is associated with dysfunctional autoregulation of cerebral blood flow; in this situation, cerebral blood flow is dependent on the arterial blood pressure. As a result, hypotension (which is common after return of spontaneous circulation) can aggravate post-arrest brain injury, and should be treated aggressively. However, the optimal pressure in the post-arrest period is unclear. Although the standard recommendation for hypotension is to maintain a mean arterial pressure (MAP) ≥65 mm Hg (see Chapter 14), observational studies show that patients with higher blood pressures in the early post-arrest period have better neurologic outcomes (43,44). Therefore, targeting an MAP that is higher than usual (e.g., ≥75 mm Hg is a reasonable consideration in the early hours following return of spontaneous circulation.

VASOPRESSOR AGENT: There is a tendency to continue using epinephrine as a vasopressor for the hypotension that follows return of spontaneous circulation, but there is evidence that norepinephrine is associated with fewer cardiovascular deaths and improved neurologic outcomes when compared to epinephrine (45).

Glycemic Control

Hyperglycemia in the early post-arrest period is associated with a poor neurologic outcome (46), so attention to glycemic control is advised. However, strict glycemic control in critically ill patients is associated with frequent episodes of hypoglycemia (47), which also promotes neurologic injury, so a higher-than-normal *range of 145–180 mg/dL* for blood glucose is considered a reasonable target of glycemic control following cardiac arrest (48). As an adjunct to this practice, it seems wise to *avoid dextrose-containing intravenous solutions* whenever possible.

Predicting Neurologic Recovery

In patients who do not regain consciousness after return of spontaneous circulation (ROSC) or targeted temperature management, the single most important determination is the likelihood of neurologic recovery. The following is a description of what is involved in this determination. The predictors of a poor outcome with a high degree of certainty (i.e., few or no false positive results) are shown in Table 21.5. (A poor outcome is defined as death, persistent coma or vegetative state, or severe disability.)

TABLE 21.5 Predictors of a Poor Neurologic Outcome with a High Degree of Certainty
1. Absence of pupillary light reflexes bilaterally on day 4 after ROSC.
2. Absence of corneal reflexes bilaterally on day 4 after ROSC.
3. Absence of oculocephalic or gag reflexes on day 2 after ROSC.
4. Myoclonic status at any time after ROSC.
5. Bilateral absence of the N20 peak on somatosensory evoked potentials.
6. An EEG that shows nonconvulsive status epilepticus or background suppression with periodic discharges.

7. A CT scan that shows diffuse cerebral edema.

From Reference 51.

Time to Awaken

Most (80–95%) patients who regain consciousness after ROSC are awake after 72 hours (49), but it can take 5 days or even longer for all patients to awaken, especially patients subjected to induced hypothermia (i.e., target temps of 32–34° C) (50). In general, the time to awaken can be a poor predictor of neurologic outcome in the first week following CPR, especially in patients who are heavily sedated in the early period following ROSC.

Clinical Examination

The only findings on clinical examination that are highly predictive of a poor outcome are the absence of brainstem reflexes. The ones that predict a poor outcome with a high degree of certainty include the absence of pupillary light reflexes bilaterally or corneal reflexes bilaterally on day 4 after ROSC, and the absence of gag or oculocephalic reflexes on day 2 after ROSC (51). Contrary to popular perception, an abnormal extensor response to pain (i.e., decerebrate posturing) is not highly predictive of a poor neurological recovery (51).

Myoclonus

The appearance of myoclonus (sudden, brief, involuntary jerks) early after ROSC is an unfavorable sign, but it does not eliminate the possibility of a neurologic recovery. However, the appearance of myoclonic status epilepticus at any time after ROSC predicts a poor outcome with almost 100% certainty (51).

Evoked Potentials

Somatosensory evoked potentials are averaged responses in the central nervous system to peripheral nerve stimulation, and are measured at the skin surface (like an EEG). Bilateral absence of the N20 peak (from the somatosensory cortex) with median nerve stimulation is highly predictive of a poor outcome (51).

Electroencephalography

The following findings on an electroencephalogram (EEG) are highly predictive of a poor neurologic outcome: 1) an isoelectric EEG (i.e., all activity <2 μV); 2) the presence of nonconvulsive status epilepticus, and 3) background suppression with superimposed periodic discharges (51).

Imaging

Computed tomography of the brain can reveal a massive stroke or intracerebral hemorrhage with herniation, both indicating very poor prognosis. CT evidence of diffuse cerebral edema is also highly predictive of a poor outcome, especially if present on day 2 or later after the cardiac arrest (51).

A FINAL WORD

Perception vs. Reality

Cardiopulmonary resuscitation has always enjoyed a popularity far greater than deserved. This is evident in surveys of the general public, where 95% of respondents have unrealistic expectations about CPR (52), including the belief that more than half of cardiac arrest victims survive and return to daily life with no residual effects (53). Television shows fuel this perception, where CPR is portrayed as a success in up to 75% of cases (54).

The reality of CPR is far removed from the public perception, as shown in Table 21.1, which indicates that CPR has a successful outcome in only 1–7% of cases.

Why is this so important? Because patients decide whether CPR is performed, so faulty perception, not reality, is dictating the delivery of CPR.

References

a. Walter Lippmann (1889–1974) was an American journalist and essayist who introduced the term *cold war*, and was one of the founders of *The New Republic* magazine.

1. Kouwenhoven WB, Ing, Jude JR, Knickerbocker GG. Closed-chest cardiac massage. JAMA 1960; 173:1064–1067.
2. Tsao CW, Aday AW, Almarzoog ZI, et al. Heart disease and stroke statistics—2023 update: A report from the American Heart Association. Circulation 2023; 147:e93–e621.

Clinical Practice Guidelines

3. Wyckoff MH, Greif R, Morley PT, et al. 2022 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science with Treatment Recommendations: Summary from the Basic Life Support; Advanced Life Support; Pediatric Life Support; Neonatal Life Support; Education, Implementation and Teams; and First Aid Task Forces. Resuscitation 2022; 208–288.
4. Panchal AR, Bartos JA, Cabañas JG, et al. Adult Basic and Advanced Life Support Writing Group. Part 3: Adult basic and advanced life support: 2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2020; 142(suppl 2):S366–S468.
5. Travers AH, Perkins GD, Berg RA, et al., on behalf of the Basic Life Support Chapter Collaborators. Part 3: Adult basic life support and automated external defibrillation. 2015 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science with Treatment Recommendations. Circulation 2015; 132(suppl 1):S51–S83.
6. Callaway CW, Soar J, Aibiki M, et al. Part 4: Advanced life support. 2015 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science with Treatment Recommendations. Circulation 2015; 132(suppl 1):S84–S145.

Instruction Manuals

7. American Heart Association. Basic Life Support Provider Manual. Dallas, TX: American Heart Association, 2016.
8. American Heart Association. Advanced Cardiovascular Life Support Provider Manual. Dallas, TX: American Heart Association, 2016.

Basic Life Support

9. Wit L, Kramer-Johansen J, Mykelbust H, et al. Quality of cardiopulmonary resuscitation during out-of-hospital cardiac arrest. JAMA 2005; 293:299–304.
10. Berg RA, Hemphill R, Abella BS, et al. Part 5: Adult basic life support: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2010; 122 (suppl 3):S685–S705.
11. Aufderheide TP, Lurie KG. Death by hyperventilation: A common and life-threatening problem during cardiopulmonary resuscitation. Crit Care Med 2004; 32(Suppl):S345–S351.
12. Abella BS, Alvarado JP, Mykelbust H, et al. Quality of cardiopulmonary resuscitation during in-hospital cardiac arrest. JAMA 2005; 293:305–310.

Advanced Life Support

13. Larsen MP, Eisenberg M, Cummins RO, Hallstrom AP. Predicting survival from out of hospital cardiac arrest: a graphic model. Ann Emerg Med 1993; 22:1652–1658.
14. Koster RW, Walker RG, Chapman FW. Recurrent ventricular fibrillation during advanced life support care of patients with prehospital cardiac arrest. Resuscitation 2008; 78:252–257.
15. Sakai T, Iwami T, Tasaki O, et al. Incidence and outcomes of out-of-hospital cardiac arrest with shock-resistant ventricular fibrillation: Data from a large population based cohort. Resuscitation 2010; 81:956–961.

16. Bartos JA, Carlson K, Carlson C, et al. Surviving refractory out-of-hospital ventricular fibrillation cardiac arrest: critical care and extracorporeal membrane oxygenation management. Resuscitation 2018; 132:47–55.
17. Herlitz J, Ekstrom L, Wennerblom B, et al. Adrenaline in out-of-hospital ventricular fibrillation. Does it make any difference? Resuscitation 1995; 29:195–201.
18. Jacobs IG, Finn JC, Jelineck GA, et al. Effect of adrenaline on survival in out-of-hospital cardiac arrest: A randomized double-blind placebo-controlled trial. Resuscitation 2011; 82:1138–1143.
19. Perkins GD, Ji C, Deakin CD, et al; and the PARAMEDIC2 Collaborators. A randomized trial of epinephrine in out-of-hospital cardiac arrest. N Engl J Med 2018; 379:711–721.
20. Sun S, Tang W, Song F, et al. The effects of epinephrine on outcomes of normothermic and therapeutic hypothermic cardiopulmonary resuscitation. Crit Care Med 2010; 38:2175–2180.

Resuscitation Monitoring

21. Rolston DM. Time is running out for manual pulse checks, as ultrasound races past. Resuscitation 2022; 179:59–60.
22. Ochoa FJ, Ramalle-Gomara E, Carpintero JM, et al. Competence of health professionals to check the carotid pulse. Resuscitation 1998; 37:173–175.
23. Kang SY, Jo IJ, Park JE, et al. Point-of-care ultrasound compression of the carotid artery for pulse determination in cardiopulmonary resuscitation. Resuscitation 2022; 179:208–213.
24. Sheak KR, Wiebe DJ, Leary M, et al. Quantitative relationship between end-tidal carbon dioxide and CPR quality during both in-hospital and out-of-hospital cardiac arrest. Resuscitation 2015; 89:149–154.
25. Kolar M, Krizmaric M, Klemen P, Grmec S. Partial pressure of end-tidal carbon dioxide predicts successful cardiopulmonary resuscitation in the field: a prospective observational study. Crit Care 2008; 12:R115.
26. Sanders AB, Kern KB, Otto CW, et al. End tidal carbon dioxide monitoring during cardio-pulmonary resuscitation. JAMA 1989; 262:1347–1351.
27. Wayne MA, Levine RL, Miller CC. Use of end tidal carbon dioxide to predict outcome in prehospital cardiac arrest. Ann Emerg Med 1995;25:762–767.
28. White RD, Goodman BW, Svoboda MA. Neurologic recovery following prolonged out-of-hospital cardiac arrest with resuscitation guided by continuous capnography. Mayo Clin Proc 2011; 86:544–548.
29. Gottlieb M, Alerhand S. Managing cardiac arrest using ultrasound. Ann Emerg Med 2023; 81:532–542.
30. Stritch School of Medicine, medical education curriculum in radiology, at www.stritch.luc.edu, accessed 11/18/2023.
31. Lalande E, Burwash-Brennan T, Burns K, et al. Is point-of-care ultrasound a reliable predictor of outcome during traumatic cardiac arrest? Resuscitation 2021; 167:128–136.
32. Hu K, Gupta N, Teran F, et al. Variability in interpretation of cardiac standstill among physician sonographers. Ann Emerg Med 2018; 71:193–198.

Post-Resuscitation Period

33. Nolan JP, Neumar RW, Adrie C, et al. Post-cardiac arrest syndrome: epidemiology, pathophysiology, and prognostication. A scientific statement from the International Liaison Committee on Resuscitation; the American Heart Association Emergency Cardovascular Care Committee; the Council on Cardiovascular Surgery and Anesthesia; the Council on Cardiopulmonary, Perioperative, and Critical Care; the Council on Clinical Cardiology; the Council on Stroke. Resuscitation 2008; 79:350–379.
34. Huet O, Dupic L, Batteux F, et al. Post-resuscitation syndrome: potential role of hydroxyl radical-induced endothelial cell damage. Crit Care Med 2011; 39:1712–1720.
35. Zeiner A, Holzer M, Sterz F, et al. Hyperthermia after cardiac arrest is associated with an unfavorable neurologic outcome. Arch Intern Med 2001; 161:2007–2012.
36. Williams GR, Spencer FC. The clinical use of hypothermia after cardiac arrest. Am Surg 1959; 148: 462–468.
37. The Hypothermia After Cardiac Arrest Study group. Mild therapeutic hypothermia to improve the neurologic outcome after cardiac arrest. N Engl J Med 2002; 346:549–556.
38. Bernard SA, Gray TW, Buist MD, et al. Treatment of comatose survivors of out-of-hospital cardiac arrest with induced hypothermia. N Engl J Med 2002; 346:557–563.
39. Nielsen N, Wettersley J, Cronberg T, et al; for the TTM Trial Investigators. Targeted temperature management at 33° C versus 36° C after cardiac arrest. N Engl J Med 2013; 369:2197.
40. Bro-Jeppesen J, Annborn M, Hassager C, et al; for the TTM Investigators. Hemodynamics and vasopressor support during targeted temperature management at 33° C versus 36° C after out-of-hospital cardiac arrest: a post hoc study of the target temperature management trial. Crit Care Med 2015; 43:318–327.
41. Dankiewicz J, Cronberg T, Lilja G, et al; for the TTM2 Investigators. Hypothermia versus normothermia after out-of-hospital cardiac arrest. N Engl J Med 2021; 384:2283–2294.
42. Roberts BW, Kilgannon JH, Hunter BR, et al. Association between early hyperoxia exposure after resuscitation from cardiac arrest and neurologic disability: prospective multicenter protocol-directed cohort study. Circulation 2018; 137:2114–2124.
43. Müllner M, Sterz F, Binder M, et al. Arterial blood pressure after human cardiac arrest and neurologic recovery. Stroke 1996; 27:59–62.
44. Kilgannon JH, Roberts BW, Jones AE, et al. Arterial blood pressure and neurologic outcome after resuscitation from cardiac

arrest. Crit Care Med 2014; 42:2083–2091.
45. Bougouin W, Slimani K, Renaudier M, et al; for the Sudden Death Expertise Center Investigators. Epinephrine versus norepinephrine in cardiac arrest patients with post-resuscitation shock. Intensive Care Med. 2022; 48(3):300–310.
46. Calle PA, Buylaert WA, Vanhaute OA. Glycemia in the post-resuscitation period. The Cerebral Resuscitation Study Group. Resuscitation 1989; 17(suppl):S181–S188.
47. Marik PE, Preiser J-C. Towards understanding tight glycemic control in the ICU. A systematic review and meta-analysis. Chest 2010; 137:544–551.
48. Peberdy MA, Callaway CW, Neumar RW, et al. Part 9: post-cardiac arrest care. 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Circulation 2010; 122(suppl 3):S768–S786.
49. Levy DE, Caronna JJ, Singer BH, et al. Predicting outcome from hypoxic ischemic coma. JAMA 1985; 253:1420–1426.
50. Lybeck A, Cronberg T, Aneman A, et al. Time to awakening after cardiac arrest and the association of targeted temperature management. Resuscitation 2018; 126:166–171.
51. Sandroni CD, Arrigo S, Cacciola S, et al. Prediction of poor neurologic outcome in comatose survivors of cardiac arrest: a systematic review. Intensive Care Med 2020; 46:1803–1851.

A Final Word

52. Jones GK, Brewer KL, Garrison HG. Public expectations of survival following cardiopulmonary resuscitation. Acad Emerg Med 2000; 7:48–53.
53. Marco CA, Larkin GL. Cardiopulmonary resuscitation: knowledge and opinions among the U.S. general public. State of the science-fiction. Resuscitation 2008; 79:490–498.
54. Diem SJ, Lantos JD, Tulsky JA. Cardiopulmonary resuscitation on television. Miracles and misinformation. N Engl J Med 1996; 334:1578–1582.

Section VIII

RESPIRATORY DISORDERS

Respiration is thus a process of combustion, in truth very slow, but otherwise exactly like that of charcoal.

Antoine Lavoisier